

**VIRTUAL PHYSIOTHERAPY GUIDER
FOR KNEE OSTEOARTHRITIS USING
COMPUTER VISION & MACHINE LEARNING**

by

H. L. Hasith Deshan.

A Report Submitted in Partial Fulfillment of the Requirements for the Degree of
Bachelor of Technology (Hons) in Mechatronics Technology

Department of Electrical and Electronics Technology

Faculty of Engineering Technology

University of Vocational Technology

Sri Lanka

February 2024

AUTHOR'S DECLARATION

I, H. L. Hasith Deshan, declare that the research work carried out for this Report was in accordance with the regulations of the University of Vocational Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the Report, including final revisions.

Name:

Student ID Number:

Signature:

H. L. Hasith Deshan

H/21/35/MEC/17/B1/33



Date: 10. 02. 2024

Supervisor's Recommendation

Name:

Designation:

Signature:

Dr. Sanjeewa Sondarangalla

Senior Lecturer



Date: 10. 02. 2024

Dr. S. D. A. Sanjeewa
Head/Senior Lecturer (G1)
Department of Electro-Mechanical Technology
Faculty of Engineering Technology
University of Vocational Technology
Ratmalana, Sri Lanka.

ACKNOWLEDGMENTS

The researcher would like to express his heartfelt gratitude to all the people who served as a great help in any means, and thus made this study a success.

First and foremost, to my research supervisor, Dr. Sanjeewa Sondarangalla, for his guidance, patience, and constant support by sharing his expertise throughout the study; for unselfishly devoting his time despite his hectic schedule as a senior lecture.

To the leading professor of our university, Prof. C. Mahesh Edirisinghe, Vice Chancellor of the University of Vocational Technology, for giving his counsel and suggestions for every activity that happens in the university. Also, to Dr. Jayalal Wetthasinghe, Dean of the Faculty of Engineering Technology, and to Eng. Dilshan Ganegoda, Head of the Department (HOD) of Electro Mechanical Technology, for their invaluable notions and words, intellectual suggestions, and their advice throughout the fourth year. And to the rest of the lecturers, academic and non-academic staff of our faculty for their tremendous efforts and teachings throughout the journey of my university life.

To the brightest woman I have had the pleasure of having in my life, Afka Alikhan, a medical student at Tver State Medical University in Russia, for generously sharing her medicinal expertise and knowledge about ‘Knee Osteoarthritis.’ Her insights have significantly contributed to all the sections of this research, including the literature review, methodology, and the results and analysis.

Last but not least, to my parents, who provided me with unending love, support, understanding, and, more importantly, financial assistance during the study. As well as to my friends who supported me all the way amidst their own involvement in their respective studies.

ABSTRACT

Knee osteoarthritis is a common condition that can cause pain, stiffness, and reduced mobility in the knee joint. Rehabilitation exercises are an important part of the treatment for knee osteoarthritis, as they can help to improve flexibility, strength, and mobility. Current clinical and medical methods for determining whether an exercise is being performed correctly by measuring the knee joint and providing feedback to the patient necessitate professional involvement as well as wearable and non-wearable tools and devices. This process entirely depends on the domain knowledge and experience of the observer. However, accessing physiotherapy and using expensive tools and devices can be difficult for some individuals due to geographic, financial, or time constraints. This study aims to address the limitations of current rehabilitation exercises observing techniques of knee osteoarthritis by reducing the subjectiveness that occurs with domain expert involvement.

Virtual physiotherapy technologies, such as those using computer vision and machine learning, have the potential to provide a more accessible and convenient alternative to traditional in-person physiotherapy. These technologies can guide individuals through rehabilitation exercises and provide real-time feedback on technique and form. This dissertation presents a virtual physiotherapy guider for individuals with knee osteoarthritis that uses skeleton point-based joint tracking to track the movement and position of the body and provide feedback to help users perform exercises correctly. The system was evaluated in a pilot study, and the results showed that it effectively improved the rehabilitation exercises' form and technique and maintained the specific exercise's correctness. The virtual physiotherapy guider has the potential to be a valuable tool for individuals with knee osteoarthritis, providing a convenient and accessible alternative to traditional physiotherapy.

CONTENT

	Page
ACKNOWLEDGMENT	IV
ABSTRACT	V
LIST OF TABLES	VIII
LIST OF FIGURES	IX
LIST OF ABBREVIATIONS	X
CHAPTER 1 INTRODUCTION	1
1.1 Background of the Study	1
1.2 Statement of the Problem	2
1.3 Research Questions	3
1.4 Project Aim and Objectives	5
1.4.1 Project Aim	5
1.4.2 Project Goals	5
1.4.3 Project Objectives	5
1.5 Justification of the Research	6
1.6 Scope and Delimitations	7
1.6.1 In scope	7
1.6.2 Out scope	7
1.7 Conclusion	7
CHAPTER 2 LITERATURE REVIEW	8
2.1 Study review of the impact of exercises on the recovery of osteoarthritis patients	8
2.2 Review of study related to exercises for knee osteoarthritis	11
2.3 Study review of implemented devices and technologies used for osteoarthritis	18
2.3.1 KSS and KOOS scores	24
2.4 The importance of guidance and assistance to improving quality, adherence and effectiveness of exercise	26
2.5 Study review of human body posture and movement tracking devices and technologies	30
2.5.1 Non-wearable sensors	31
2.5.2 Transfer learning technology	35

2.6 Body posture detection system used in sports and medical domains	36
2.7 Chapter Summary	39
CHAPTER 3 METHODOLOGY	41
3.1 Introduction	41
3.1.1 The significant points when choosing a skeleton tracking technology to measure the Knee Angle	44
3.1.2 Identify clinically accepted rehabilitation exercises for knee osteoarthritis	45
3.1.3 Selecting the suitable skeleton tracking technology	47
3.1.4 Selection of the MediaPipe Framework for Skeletal Point Extraction	48
3.1.5 Key points used for the features of the model using MediaPipe Framework	49
3.1.6 Identifying clinically accepted parameters for Knee Range of Motion	50
3.1.7 Calculating the Euclidean distance between two points in a human skeleton using MediaPipe	55
3.1.8 Calculating the angle between three points in a human skeleton using MediaPipe	56
3.1.9 Optimal camera angle for accurate detection of Knee Joint Angle in patients	59
3.1.10 Biomechanical model development	60
3.1.11 Performance parameter model development	61
3.1.12 Model Evaluation	63
CHAPTER 4 RESULTS AND ANALYSIS	65
4.1 Introduction	65
4.2 Patient Inclusion Criteria for Virtual Physiotherapy Guider Study in Knee Osteoarthritis	65
4.2.1 Patient Inclusion Criteria	65
4.2.2 Patient Exclusion Criteria	65
4.3 Randomization	66
4.4 Intervention	68

4.5 Guidelines for the relevant exercises	68
4.5.1 Straight Leg Raise	68
4.5.2 Knee Extension	68
4.5.3 Reverse Lunge	69
4.6 Outcome measurements	70
4.6.1 Calculate the KOOS score	70
4.6.2 Scoring and Interpretation	70
4.6.3 KOOS survey for the Knee Osteoarthritis	71
4.7 Patients' Demographics table	75
4.8 Results	76
4.9 Discussion	79
CHAPTER 5 CONCLUSION AND RECOMMENDATION	81
5.1 Introduction	81
5.2 Conclusion	81
5.3 Limitations and Constraints	82
5.4 Future enhancements for further research	83
REFERENCES	86
APPENDICES	99

LIST OF TABLES

Tables	Page
Table 2.1 Types of exercise selected which will be guided by the virtual assistant	17
Table 2.2 Summary of the accuracy of the improper exercise	19
Table 2.3 Table for converting raw summed scores to interval level scores from 0 (total knee disability) to 100 (perfect knee health)	25
Table 3.1 Selected exercises related to the levels of the physical therapy plan	46
Table 3.2 Comparison Matrix	47
Table 4.1 KOOS score of patients before intervention taken on 20/12/2023	74
Table 4.2 Demographics table of the participating patients	75
Table 4.3 KOOS score of patients after intervention taken on 17/01/2024	76
Table 4.4 (Table 1) Outcomes of Group – A, before intervention and after 4 weeks of the Virtual physiotherapy guider - directed exercise	76
Table 4.5 (Table 2) Outcomes of Group – B, before intervention and after 4 weeks of the handout self - directed exercise	77
Table 4.6 (Table 3) Comparison of outcomes between the two groups after 4 weeks	77
Table 4.7 (Table 4) Comparison of outcome exercise between the two groups	77

LIST OF FIGURES

Figures	Page
Figure 2.1 Exercises that stretching and strengthening the muscles around the knee	12
Figure 2.2 Patient exercise program	13
Figure 2.3 SAE and SLR exercises	18
Figure 2.4 placement of the sensors around the leg	19
Figure 2.5 Movement biofeedback on toe out angle during walking gait	21
Figure 2.6 Feature extraction process of A-SKE posture	22
Figure 2.7 Sagittal plane view of the elliptical trainer	24
Figure 2.8 Reliability of all subscales of the KOOS (Groot et al., 2008)	26
Figure 2.9 Robot platform and the exercise setup	29
Figure 2.10 3D depth sensing cameras	32
Figure 2.11 Framework of the MobiFit	33
Figure 2.12 Feature points of human motion behaviour characteristics	36
Figure 3.1 High Level architecture of the research design	41
Figure 3.2 Six (6) Landmarks (blue colored) mainly focused on this research	50
Figure 3.3 The external and internal rotations of the leg when the knee is bent	52
Figure 3.4 Specific landmarks on the leg to measure the knee range of motion	54
Figure 3.5 Measuring knee range of motion using a goniometer	54
Figure 3.6 Euclidean distance on a 2D plane	56
Figure 3.7 A triangle depicting the Law of Cosines	57
Figure 3.8 Optimal camera angle for accurate detection of Knee Joint Angle	60
Figure 4.1 The pathway of each group throughout the study protocol	67

LIST OF ABBREVIATIONS

OA	= Osteoarthritis
WHO	= World Health Organization
HEP	= Home Exercise Programs
VR	= Virtual Reality
ROM	= Range Of Motion
PTs	= Physical Therapists
KOA	= Knee Osteoarthritis
IG	= Intervention Group
CG	= Control Group
HBEI	= Home-Based Exercise Intervention
ACR	= American College of Rheumatology
EULAR	= European League Against Rheumatism
OASRI	= Osteoarthritis Research Society International
NSAID _s	= Non-Steroid Anti-Inflammatory Drugs
VAS	= Visual Analogue Scale
KOOS	= Knee And Osteoarthritis Outcome Score
LED	= Light Emitting Diode
SLR	= Straight Leg Raise
SAE	= Short Arc Exercise
A-SKE	= Active Seated Knee Extension
A-SHS	= Active Seated Hamstring Stretch
A-HSKE	= Active Heel Slide Knee Extension
A-PKF	= Active Prone Knee Flexion
AAPKF	= Actively Assisted Prone Knee Flexion
EKAM	= External Knee Adduction Moment
KSS	= Knee Society Score
IMUs	= Inertial Measurement Units
AI	= Artificial Intelligence
3D	= Three Dimensional
AACN	= American Association of Colleges of Nursing
SAR	= Socially Assistive Robot

sEMG	= Surface Electromyography
SVM	= Support Vector Machine
2D	= Two Dimensional
AOI	= Angle Of Inclination
RMSE	= Root Mean Squared Error
MAE	= Mean Absolute Error
LSTM	= Long Short-Term Memory System
RFID	= Radio-frequency identification
GRFs	= Ground Reaction Force
PFPS	= Patellofemoral pain syndrome
SVM	= Support Vector Machine
MOCAP	= Motion Capture
ML	= Machine Learning
HPE	= Human Pose Estimation

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Osteoarthritis (OA), also known as degenerative joint disease, is a leading cause of pain and disability and a source of socioeconomic costs among individuals over 40 years old worldwide. Currently, OA has been listed as the fastest-growing major health condition, and the World Health Organization (WHO) has classified it as the second leading cause of disability (Li et al., 2022). Guidelines from the American College of Rheumatology (National Center for Biotechnology Information, 2022) and the European League Against Rheumatism (Pendleton, 2000), although generated in 2000, are consistent in recommending aerobic exercise, muscle strengthening, and preservation of joint mobility as central components of the nonpharmacologic approach to the treatment of knee OA.

The 2007 recommendations of the Osteoarthritis Research Society International (Zhang et al., 2008) are more inclusive of the current literature and again stress the value of exercise, whether water-based, aerobic, muscle-strengthening, or physiotherapy-based. Further, these guidelines stress that there are no statistical differences in the benefits derived from pharmacologic and nonpharmacologic modalities in the treatment of knee OA (Zhang et al., 2008). Despite these findings, in a Canadian study at a teaching clinic (DeHaan, 2022), was rarely able to find documentation of the use of nonpharmacologic therapy. However, based on the evidence, judicious use of exercise is a compelling modality in the treatment of knee OA, and this intervention incurs little risk.

In recent years, there has been a shift from primarily pharmacologic therapy to non-pharmacologic therapy, owing to the limited benefits of the former and evidence that non-pharmacologic approaches are more likely to relieve symptoms in the long term and to delay or prevent functional decline. Home exercise programs (HEP) serve as a crucial complement to outpatient rehabilitation therapy, as they save the cost of supervised physical therapy sessions (Makarm et al., 2021) while also offering a high level of treatment to obtain the benefits of exercise. It is important that exercise should be practised in the correct postures and forms. Incorrect postures can cause

unproductive results to the extent of pernicious effects. This creates the necessity for a physiotherapist to administer the appropriate individual posture.

The availability of physiotherapists is very limited, and that makes it a costly affair as well. Thus, the development of a self - coaching system with the help of machine learning could help individuals evaluate their own exercise performance without professional assistance. Our objective is to develop an easy-to-use interface technology that assists users who need to perform workouts to speed up their recovery or healing without needing active professional supervision.

Several studies aim to identify the accuracy of human body postures using motion analysis. Introducing low-cost depth sensors like Microsoft Kinect, Orbbec Astra Pro and pose estimation models such as the MoveNet model have made human movement tracking easier. Currently, several studies have been conducted using inertial sensors for motion analysis. Still, most measure and recognize the human movement and postures for one part of the body where the sensor is attached (Cuesta-Vargas et al., 2010). To be used in rehabilitation exercise consists of the repetition of a movement while maintaining a specific posture of the whole body. Hence, most current products are not suitable for assessing the exercise posture of the whole body.

In recent years, skeleton detection approaches have attracted the research community's attention and have become increasingly popular for action recognition. In contrast to video-based approaches, which still contain noise from the video background, skeleton data only provides the position coordinates of joints, avoiding the effects of lighting and appearance and the effects of the background. This study intends to build a computational model using the skeleton joint variation of patients, which involves assisting in exercise by giving feedback to correct and improve the body's position and record the overall improvement and recovery of the disease.

1.2 Statement of the Problem

At present, there is no proper technology-based assistance for the rehabilitation process of osteoarthritis in the health sector. Personal trainers and Physiotherapists are the ones that monitor manually. This is a disadvantage because having personal assistance throughout the rehabilitation process to correct and give feedback for the exercise performed is not practical for all individuals affected with OA. Although accessing

physiotherapy and using expensive tools and devices can be difficult for some individuals due to geographic, financial, or time constraints. Also, manually with the help of professionals, couldn't detect accurately relevant tools and equipment whether the patient performed the exercise correctly, but with the help of AI, can detect the angle between joints and give feedback to correct the position, which will be effective in the treatment of the disease.

And there's no existing literature on models that combine following all detection of the body position, knee angle and also the distance between two legs during exercise and give real-time feedback to the patient. In order to fulfil all that mentioned above, full body tracking is needed. To provide near real-time comments to the patients involved in the exercise, point cloud data of joint movement captured from a suitable skeleton tracking technology will be used in this study. Since domain experts are involved in patient assessment and use clinical and theoretical standards during the examination, it is important to identify how those clinical measurements can be mapped with a cloud of skeleton data. In this study, this problem will be addressed, and the proposed framework will be explained more in the following sections.

1.3 Research Questions

The main research hypothesis is that rehabilitation exercises for knee osteoarthritis can be done effectively with the help of a virtual assistant which identifies the position of the body with the help of main skeleton points variation and gives real-time feedback to correct it for effective and quality treatment of the disease and improve day to day activities by having a record of improvement.

Here are some potential research questions that could be explored in relation to the main research hypothesis:

- 1. How effective is the virtual assistant at identifying the position of the body and providing real-time feedback to correct it during rehabilitation exercises for knee osteoarthritis?**

The virtual assistant can assist in identifying the body's position and movements, providing real-time feedback, recording important data for rehabilitation exercises, and presenting with the patient throughout the workout session, which is considered a gold standard of virtual over manual assistant. Some technologies and approaches, such as

virtual reality (VR) systems, wearable sensors, and mobile apps, can help with rehabilitation exercises. Nevertheless, there are limitations to using these tools; for example, wearable sensors detect only the part where the sensor is worn and wearing it causes some discomfort to the patient. For rehabilitation exercises, it is necessary to detect whole-body movements and also it should be cost-effective. However, when it comes to knee osteoarthritis, it is critical to identify the body position and movements accurately.

The virtual assistant's effectiveness would be determined by several factors, including the specific technologies used to track body position and provide feedback, the accuracy and reliability of these technologies, and the design and implementation of the rehabilitation exercises. Therefore, a reliable, conveniently accessible and fast human skeleton point-based motion tracking technology should be selected.

2. What rehabilitation exercises currently available for knee osteoarthritis can be analysed through this human skeleton point-based motion tracking technology?

Range of Motion (ROM) exercises such as straight leg raises, bent leg raises, and knee bends to improve the flexibility and mobility of the knee joint, as well as some strengthening exercises, can also be analysed using this technology. However, it is essential to discuss this with a domain expert to identify the most appropriate exercises for our specific needs and goals.

3. What is the clinically accepted method to identify the knee angle of a patient, and how to create criteria to map the accepted measurements with the output of the skeleton points detection technique in computer vision?

The clinically accepted method for measuring knee angle is to use a goniometer. A *goniometer* is a device consisting of two arms connected by a hinge. One arm of the goniometer is placed along the thigh, and the other is along the lower leg. The angle between the two arms is then measured and recorded as the knee angle (Pereira et al., 2015).

To map the accepted measurements with the output of a skeleton points detection technique in computer vision, we should create criteria based on the positions of the

detected skeleton points. For example, we should define the positions of the points corresponding to the hip, knee, and ankle on the left and right sides of the body and use these positions to calculate the knee angle using trigonometry. After that, a couple of arbitrary angles can be taken, and the measurements of those angles can be compared using both methods. With the collected data we can create criteria for the stage of disease based on forming a connection between capable knee angle and bone stiffness. This will be the parameter that we use to form the result of the treatment by comparing it before and after exercising with the help of a virtual assistant.

1.4 Project Aim and Objectives

1.4.1 Project Aim

The aim of this project is to build a virtual assistance to guide patients to perform the exercises and movements accurately, making them adhere to exercises, to have records of improvements and thereby improve their quality of life.

1.4.2 Project Goals

The goal of the project is to replace Physical therapists (PTs), who carefully assess joint motion, muscle strength and endurance, and performance of activities of daily living, by designing a Virtual Physiotherapy Guider for Knee Osteoarthritis using Computer Vision & Machine Learning which may play an important role in helping these individuals become more physically active.

1.4.3 Project Objectives

The primary objective of this study is to reduce the cost of physiotherapy; by providing a virtual alternative to in-person physiotherapy, the guide could help reduce the cost of treatment while also providing an easy interface to ensure that patients adhere to exercise to achieve adequate improvement. In order to accomplish this, the secondary objectives listed below must be met.

- Determining suitable existing exercises for knee osteoarthritis that can be analysed using the skeleton point-based motion tracking technology.
- Determining a suitable and cost-effective non-wearable motion tracking technology to conduct movement analysis.

- Developing suitable criteria to map clinical measurements of identifying the knee angle of a patient using the skeleton point-based motion tracking technology.
- Evaluate the patient's condition, and have a customized exercise plan according to age and stage of the disease.
- Develop a user-friendly interface by providing audio or visual instructions to help users speed up their recovery or healing without wearing additional sensors or devices.
- Give real-time corrective feedback when exercises are performed incorrectly and record essential data afterwards.
- Evaluate the accuracy and reliability of the final model using an expert in the medical domain.

1.5 Justification of the Research

Physical activity is essential to optimizing both physical and mental health and can play a vital role in the management of arthritis. Regular physical activity can keep the muscles around affected joints strong, decrease bone loss and may help control joint swelling and pain. Regular activity replenishes lubrication to the cartilage of the joint and reduces stiffness and pain (Zeng et al., 2021). Exercise also helps to enhance energy and stamina by decreasing fatigue and improving sleep (U.S. Department of Health and Human Services, 2023). Exercise can enhance weight loss and promote long-term weight management in those with arthritis who are overweight. Exercise may offer additional benefits to improving or modifying arthritis (Minor, 1999)- (Susan Bartlett, 2018). But it should be guided to get the best result and prevent disastrous results which lead to accelerated symptoms of the disease (Hunter & Eckstein, 2009).

As mentioned in the earlier sections, various approaches have been introduced for pose detection, in various field but this piece of work focus to apply the human skeleton tracking technology on health sector which can substitute manual physical therapist, to guide and assist rehabilitation exercise for knee osteoarthritis patients by giving real-time corrective feedback. If there's a solution which can replace the physiotherapist and gain the complete benefit with less use of resources (such as domain experts and

expensive equipment), it would be a contribution to effective treatment at home for individuals without an obstacle. It will be a great innovation in health sphere because, OA has been listed as the fastest-growing major health condition. The World Health Organization (WHO) has classified it as the second leading cause of disability (Li et al., 2022).

1.6 Scope and Delimitations

1.6.1 In scope

- Exploring body tracking technologies that provide accurate joint data values when performing the exercise.
- Development of a performance parameter model trained by a specialist in the field of physiotherapy, which produces accurate position of the body while performing the exercise which is selected according to the need.
- Development of a biomechanical model to detect whether the patient is performing the exercise correctly.
- Patient's lateral view is tracked using a skeleton tracking approach.

1.6.2 Out scope

- The biomechanical model developed in this study can be used only to guide the exercises (specific exercises according to age and condition of the patient), which the system has trained not for all category of OA patients.
- A healthy individual trains the performance parameter model; therefore, the patient cannot achieve the exact pose. So, a range of joint angles should be considered correct, and improvement is calculated based on reaching the perfect pose by the patient.
- Improvements with use of another camera to capture patient's anterior view.

1.7 Conclusion

This chapter laid the foundations for the dissertation. It introduced the research problem and research questions and hypotheses. Then the research was Justified, definitions were presented, the methodology was briefly described and justified, and the limitations were given. On these foundations, the dissertation can proceed with a detailed description of the research.

CHAPTER 2

LITERATURE REVIEW

This section is dedicated to presenting the literature on knee osteoarthritis and related technologies based on previous research. Section 2.1 describes the study review of the impact of exercises on the recovery of osteoarthritis patients. Section 2.2 contains the exercises currently used in the medical domain for knee osteoarthritis, and the exercises chosen to be analysed by this model are explained in detail. Section 2.3 explains the Study review of implemented devices and technologies used for osteoarthritis. The significance of guidance and assistance in improving exercise quality is discussed in Section 2.4, which is concerned with researchers' previously published exercise guiding systems. Sections 2.5 and 2.6 describe human body posture and movement tracking devices and technologies and body posture detection systems used in sports and the medical domains. This section also discusses the skeleton point-based biomechanical models that have been proposed for pose recognition, pose classification, and pose estimation. The summary can be found in Section 2.7.

2.1 Study review of the impact of exercises on the recovery of osteoarthritis patients.

Osteoarthritis (OA) is the most common form of arthritis. Some people call it a degenerative joint disease or “wear and tear” arthritis (Centers for Disease Control and Prevention, 2022). It is most common in the hands, hips, and knees. With OA, the cartilage within a joint begins to deteriorate, and the underlying bone changes. Initially, researchers thought of OA as a single articular cartilage disease. Although cartilage loss is a prominent feature of OA, contemporary models recognize that the entire joint organ is affected by OA (Hunter & Eckstein, 2009). These alterations commonly occur gradually and worsen with time. Pain, stiffness, and oedema are all symptoms of OA. In certain circumstances, it also decreases function and incapacity; some people cannot accomplish their everyday work and activities. The signs and symptoms of OA are as follows.

- Pain or aching
- Stiffness
- Decreased range of motion (or flexibility)

- Swelling

Exercise is vital for arthritis patients to avoid the symptoms indicated above. It improves strength and flexibility, alleviates joint pain, and helps to relieve fatigue. Exercise can help to increase health and fitness without affecting the joints of arthritis patients. With the support of a patient's current treatment programme, exercise can:

- Reduces joint stiffness
- Increases flexibility and endurance
- Improves mood and self-esteem
- Aids in sleep
- Maintains weight control
- Provides more energy

In addition to relieving osteoarthritis pain, exercise can also help with other health issues such as osteoporosis, diabetes, and heart disease. The previous study (Chen et al., 2019) has examined the significance of home-based exercise in the treatment of knee osteoarthritis. Knee osteoarthritis (KOA) is a prevalent condition among the elderly that causes pain, loss of physical function, and impairment. According to this study, the prevalence of symptomatic KOA in the United States is estimated to be 7-33%, with the frequency increasing with age; the knee is the most common site of symptomatic OA (Lawrence et al., 2007). Because there is no known cure for OA, non-pharmacological treatments are crucial for controlling its progression, relieving symptoms, and improving knee function and quality of life (Bennell & Hinman, 2011). To observe this, the researchers took 171 senior patients (60 years of age or older) with KOA recruited from four community centers. Patients from two community centers were randomly assigned to the intervention group (IG), while patients from the other two were randomly assigned to the control group (CG). The IG patients received a 12-week HBEI as well as health education, whereas patients in the CG only received health education. In the end, they discovered that the IG improved significantly more on both measures than the CG.

Another study (Peilei, 2021) of individuals with knee osteoarthritis concluded that exercise improved symptoms and quality of life more than nonsteroidal anti-inflammatory medications and COX-2 inhibitors over a 12-week follow-up period. This

experiment comprised 166 individuals with knee osteoarthritis who were randomly allocated into groups. The experimental group was subjected to systematic exercise rehabilitation, whereas the control group was given naproxen (n = 28), diclofenac (n = 27), or celecoxib (n = 19) (medicines). The researchers compared the improvement in symptoms, knee function, and quality of life. For data analysis, the researchers used SPSS Statistics 24.0. In the results and analysis, the researchers mentioned that there are no statistically significant differences between the groups seen in age, body mass index, and sex ($P > 0.05$). The authors recorded the average Western Ontario and MacMaster Universities (WOMAC) scores, Lysholm scores, SF-36 (Short Form-36 Survey) scores, and average ranges of knee motion after treatment. They found some statistical differences between the groups. Finally, they discovered that exercise improved symptoms and quality of life in patients with knee osteoarthritis better than nonsteroidal anti-inflammatory medications and COX-2 inhibitors over a 12-week period.

The treatment of osteoarthritis (OA) comprises non-pharmacological and pharmacological techniques. Rehabilitation is extensively recommended in national and international guidelines for addressing osteoarthritis (OA) in primary care settings (Fernandes et al., 2013), (Hochberg et al., 2012), (McAlindon et al., 2014). Rehabilitation is even considered the primary treatment of OA and is recommended for all patients, according to the 2014 OA Research Society International (OARSI) recommendations. A previous review (Nguyen et al., 2016) thoroughly examined rehabilitation (exercise and strength training) and osteoarthritis. The authors conducted a critical narrative review of the efficacy and safety of OA rehabilitation and discussed evidence-based international recommendations. The articles were chosen based on the authors' expertise, self-awareness, and reflective practice. This review concentrated on land-and water-based exercise and strength training for knee, hip, and hand OA.

American College of Rheumatology (ACR), European League Against Rheumatism (EULAR) and Osteoarthritis Research Society International (OARSI) international guidelines were used to develop this review the authors. Also, the MEDLINE was searched via PubMed from inception to December 2015 for additional guidelines, trials, systematic reviews and meta-analyses with the MeSH terms “exercises” and “knee osteoarthritis.” Other aspects of OA rehabilitation are covered in this special issue.

Finally, they discovered that the efficacy of exercises and strength training for OA strongly suggests that exercise therapy be widely recommended in managing knee, hip, and hand OA. The level of evidence, however, varies by OA location. Overall, evidence suggests that exercise therapy and specific strengthening or strength training for the lower limb reduce pain and improve physical function in people with knee OA.

Exercise can help people with osteoarthritis, but some myths about its safety exist. Numerous programs teach people with osteoarthritis how to exercise safely. These programs may slightly improve pain scores, self-belief, and social function, but participants also report the myth that discomfort while exercising indicates ongoing harm. The previous analysis (nihr_wp, 2018) reviewed combines evidence from 21 randomized controlled trials evaluating exercise in hip or knee osteoarthritis and 12 studies in which patients receiving the intervention were asked about the influence of exercise on their illness. Participants were men and women over the age of 45 years. In participant interviews, it was discovered that pain, joint stiffness, tiredness, other illnesses, and people's perceptions of their physical fitness limited the type and amount of exercise they could do. Exercise pain was frequently thought to be causing additional joint damage, so people avoided activity for fear of causing further harm. But finally, participants stated that precise instructions from healthcare professionals explaining what exercises to do, what to avoid, and what to expect throughout the exercise helped reassure them that exercise is safe and helpful.

2.2 Review of study related to exercises for knee osteoarthritis.

With OA, the cartilage within a joint begins to deteriorate, and the underlying bone begins to change. These changes typically occur gradually and worsen over time. OA can cause pain, stiffness, and swelling. In some cases, it also results in reduced function and disability; some people can no longer perform daily tasks or work (Centers for Disease Control and Prevention, 2023). Since OA is a degenerative disease, it cannot be reversed; however, we can improve our quality of life by leading an active lifestyle. There is no cure for osteoarthritis, but the condition does not necessarily worsen over time. There are several treatments available to help relieve the symptoms.

The main treatments for osteoarthritis symptoms include:

- lifestyle measures - such as maintaining a healthy weight and exercising regularly

- medication - to relieve pain
- supportive therapies - to help make daily activities easier

Surgery to repair, strengthen, or replace damaged joints may be considered in a few cases where other treatments have declined.

In an article (Nejati et al., 2015), there was a single-blind RCT study with a 12-month follow-up. Fifty-six patients with knee OA were assigned into two random groups. The patients in the exercise group received exercise for knee muscles (6 exercises, stretching and strengthening the muscles around the knee) combined with non-steroid anti-inflammatory drugs (NSAIDs) and ten sessions of acupuncture and physiotherapy modalities. The non-exercise group received similar treatments except exercise program. Visual analogue scale (VAS), knee and osteoarthritis outcome score (KOOS) questionnaire, and functional tests (4 steps, 5 sit up, and 6 min walk test) were used to evaluate the patient's pain and functional status before and after treatment (1 and 3 months after intervention) and one year later at the follow-up.

This study concludes that adding exercise to other noninvasive methods that routinely are used for knee OA have boosting effect in relieving pain and improving knee function. According to this trial, combination therapy has the most effect on knee OA. Exercise therapy can be recommended for patients with even severe arthritis as an effective method combined with other palliative methods in knee OA. The exercises which were done in this study as follows (Figure 2.1).

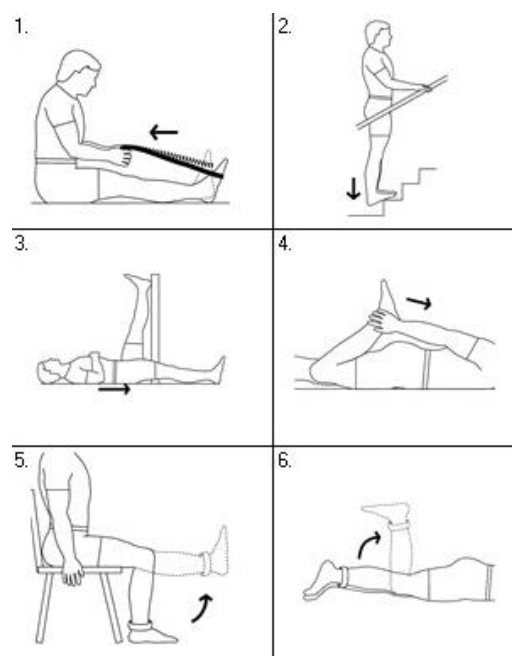


Figure 2.1 Exercises that stretching and strengthening the muscles around the knee.

In (CreakyStaff, 2019), to evaluate the effectiveness of physical therapy for osteoarthritis of the knee, 83 patients with osteoarthritis of the knee who were randomly assigned to receive treatment (n = 42; 15 men and 27 women [mean age, 60 ± 11 years]) or placebo (n = 41; 19 men and 22 women [mean age, 62 ± 10 years]). The treatment group received manual therapy to the knee, lumbar spine, hip, and ankle as needed and performed a standardized knee exercise programme in the clinic and at home. This program consisted of active range-of-motion exercises for the knee, muscle strengthening exercises for the hip and knee, muscle stretching for the lower limbs, and riding a stationary bike. Figure 2.2 depicts the patient exercise program of this study. The placebo group had a subtherapeutic knee ultrasound at an intensity of 0.1 W/cm² with a 10% pulsed mode. Both groups were treated at the clinic twice weekly for four weeks. The treatment group All activities were mutually reinforcing, with repeated gentle challenges to the end ranges of movement.

Stretching Exercises	
Exercise	Measures
1. Standing calf stretch.	Three repetitions with 30-sec hold.
2. Supine hamstring stretch.	Three repetitions with 30-sec hold.
3. Prone quadriceps stretch.	Three repetitions with 30-sec hold.

Range-of-Motion Exercises	
Exercise	Measures
1. In long sitting position, knee mid-flexion to end-range extension.	Two 30-sec bouts with 3-sec hold at end range.
2. In long sitting position, knee mid-flexion to end-range flexion.	Two 30-sec bouts with 3-sec hold at end range.
3. Stationary bike.	Five min. Increase time as tolerated.

Strengthening Exercises	
Exercise	Measures*
1. Static quad sets in knee extension.	One set of 10 repetitions with 6-sec hold, 10-sec rest between repetitions.
2. Closed-chain progression†:	
A. Standing terminal knee extension.	One 30-sec bout. Increase resistance as tolerated.
B. Seated leg press.	One 30-sec bout. Increase resistance as tolerated.
C. Dips weight-lessened.	One 30-sec bout. 1 sec during concentric phase and 2 sec during eccentric phase. Progress from 2 legs to 1 leg as tolerated.
D. Step-ups.	One 30-sec bout. Increase step height as tolerated.

Figure 2.2 Patient exercise program.

According to the findings of this study, a combination of manual physical therapy and supervised exercise provides functional benefits for patients with knee osteoarthritis. It may delay or prevent the need for surgical intervention.

Thereby We can categorize the exercises for OA into 3 groups as,

- **Strengthening exercise:** The program includes exercises beneficial for people with knee osteoarthritis. The exercises aim to improve the strength of specific muscles that will provide better support for the knee during walking and other tasks. Stronger muscles help to cushion as well as support and protect the joint. Stronger muscles also make it easier to perform many daily activities (Deyle et al., 2000).
- **Stretching exercises:** particularly benefits those with arthritis by lubricating joints and enhancing and maintaining range of motion (Ruane, 2011).
- **Flexibility exercises:** Because knee OA often makes it hard to move, flexibility exercises are essential. Doing them regularly can help increase range of motion, make the knees more flexible, and restore normal knee joint function (Arthritis Foundation, n.d.).

These exercises are essential because they help remove the knee's strain. This work intends to prove that performing exercise correctly and continuously improves the patient's lifestyle because exercise is considered one of the most important treatments for people with osteoarthritis, regardless of age or fitness level. Patients' physical activity should include a combination of exercises to strengthen their muscles and exercises to improve their general fitness (NHS Choices, 2019). Since the improvement process is slow and steady, it is necessary to adhere to physical therapy. Activities that maintain flexibility, muscle strength, and coordination, protect the cartilaginous surfaces and help maintain joint function are recommended (Sandmeier, 2000). The patients should meet our inclusion criteria, and then they were offered the opportunity to participate in the virtual rehabilitation process.

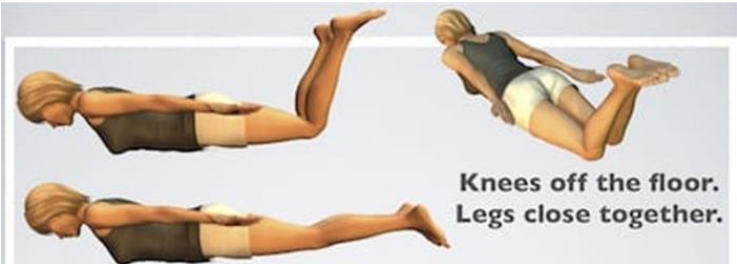
Patients are eligible if they have had no surgical procedure on either lower extremity in the past six months and have no physical impairment unrelated to the knee (no other complications or contradictions to exercise). Patients had to have sufficient English-language skills to comprehend all explanations and to complete the assessment tools.

This project aims to create a virtual system to assist exercise, which can be later tailored according to patient's needs and wants. Here we intend to analyse the results of physical therapy after performing the exercise with the virtual assistance created, based on the improvement of range of motion and function and pain assessment questionnaire.

Based on the study review above, we selected 8 exercises (Table 1) in which our virtual system guide with the use of skeleton point technology and the plan of physical therapy is shown in Figure 2.2.

Type of Exercise	Exercises
------------------	-----------

Stretching exercises Prone quadriceps stretch



Inner range quadriceps



Supine hamstring stretch



Dynamic stretching for hamstrings

Strengthening
exercises

Straight leg raise



Standing hamstring curls



Flexibility exercises Knee extension



Reverse lunge

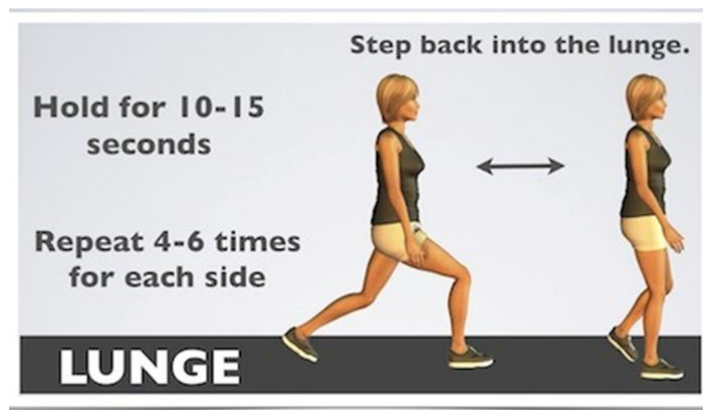


Table 2.1 Types of exercise selected which will be guided by the virtual assistant

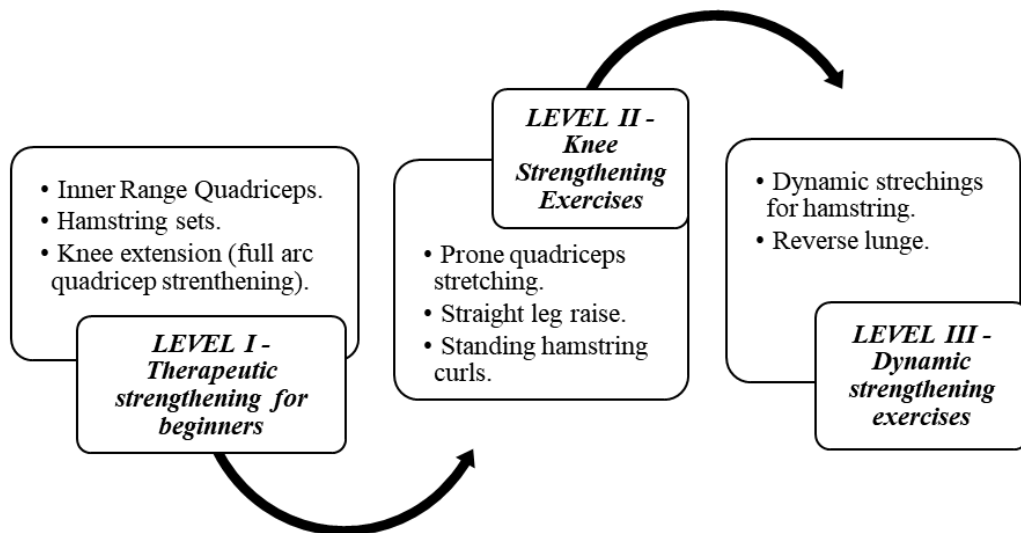


Figure 2.2 Plan of physical therapy

2.3 Study review of implemented devices and technologies used for osteoarthritis.

Modern technology and equipment are also used in osteoarthritis treatment and exercises. Individuals with OA are typically treated with prescribed physical activity in the hospital, using sophisticated and costly rehabilitation equipment and under the supervision of a physician. Another option is to do rehabilitation exercises at home. On the other hand, poor exercise movement at home may cause more severe harm to the patient. As a result, the innovators developed various technological devices such as monitoring at-home exercises and guiding the patient.

A previous study (Ishak & Addi, 2017) proposed a rehabilitation exercise monitoring device for knee OA. Two accelerometers are used in this suggested system to collect data on body motion from the thigh and shank. The Arduino is used to process the signal obtained from the accelerometers for exercise classification and detection of erroneous exercise movement using LEDs as indicators. To test the detection accuracy of the system, the subject performed two types of exercises: Short Arc Exercise (SAE) and Straight Leg Raise (SLR). Figure. 2.3 shows that SAE and SLR exercise respectively. The authors used two-node sensors consisting of the MPU6050 sensor module, which includes an accelerometer and a gyroscope, to acquire motion data from the body's movement.



Figure 2.3 SAE and SLR exercises

The placement of the sensors around the subject is shown in Figure 2.4. The accelerometer measures the human body's inclination, acceleration, gravity, and the flexion angle of a bending motion. On the other hand, the gyroscope is used to extract

information about the body's orientation. By observing the switching behaviours of the LED (ON or OFF) based on the subject's movement, the accuracy for two parameters, exercise classification and improper exercise movement, were determined. The average angle for each exercise performed by the subjects was also determined.

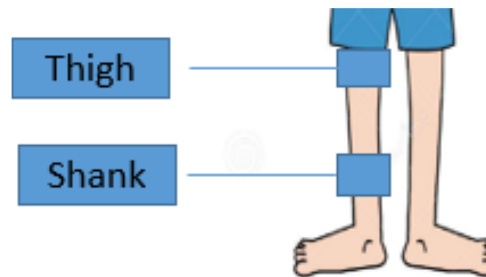


Figure 2.4 placement of the sensors around the leg

Each subject was asked to complete three (3) sets of each of the SAE and SLR exercises, with each set being repeated ten (10) times. The signal from sensor S2, located on the thigh, is the most important because it determines exercise classification. The input from sensor S2 is used as an indicator to distinguish between SAE and SLR because SAE movements only involve the shank area, which is where sensor S1 is located (Ishak & Addi, 2017). In contrast, SLR movements involve signals from both sensors. However, to determine the classification, only the data from the S2 sensor was taken to differentiate between SLR and SAE, as a signal to SAE only involves signals from S1 at the shank area.

They collected the sensors' output signals and the LED marking in the results and analysis to determine the accuracy of the incorrect SAE and SLR exercise movements. The table below (Table 2.2) summarizes the accuracy of improper exercise movement in all sets.

Exercise	Labels	Subject A	Subject B
Short Leg Arc Exercise (SAE)	Initial knee flexion > 25 ⁰ and knee not fully extended	100%	100%
Short Leg Raise Exercise (SLR)	Knee not fully extended	57%	100%
	Hip joint external rotation	57%	100%

Raise angle not approximately 45°	100%	100%
--------------------------------------	------	------

Table 2.2 Summary of the accuracy of the improper exercise

They used subjects A and B to make the comparison. When performing SAE, the results demonstrated that the system could provide 100% accuracy for both subjects. Subject A has the lowest accuracy for SLR with the error label of hip joint external rotation and knee not fully extended. This could be due to sensor instabilities. Simultaneously, the sampling frequency chosen may be one of the factors contributing to the problem (Ishak & Addi, 2017). According to the authors, it can be improved by performing proper sensor calibration. However, this system can classify different types of exercises and determine improper exercise movements by providing the average values of the key parameters.

Several walking gait modifications have been identified that reduce the load on the knee joint, which is especially important in knee osteoarthritis (OA). Researchers previously developed a Real-time movement biofeedback system for walking gait adjustment in knee osteoarthritis (Wrigley et al., 2009). This application aims for patients with knee osteoarthritis to help teach and learn gait modification measures to lessen knee loading. The proposed system uses real-time data streamed into MATLAB using Vicon Nexus software linked to a Vicon MX 8-camera motion capture system and three force platforms (AMTI, Mass., USA) installed in a ten-meter walkway (Wrigley et al., 2009).

They used Nexus software to communicate with camera motion capture in real-time and process them to the MATLAB architecture. Researchers created this method to provide movement biofeedback on the toe during the walking motion. The Vicon MX camera system tracks the 3D locations of the retroreflective markers in real-time. MATLAB uses the Tarsus protocol to extract the XYZ marker locations of the heel and toe markers and compute the angles of the left and right heel-toe vectors for a display to the subject on a rear projection screen. The following figure (Figure 2.5) shows an example of the use of the system for providing movement biofeedback on the toe-out angle during walking gait.

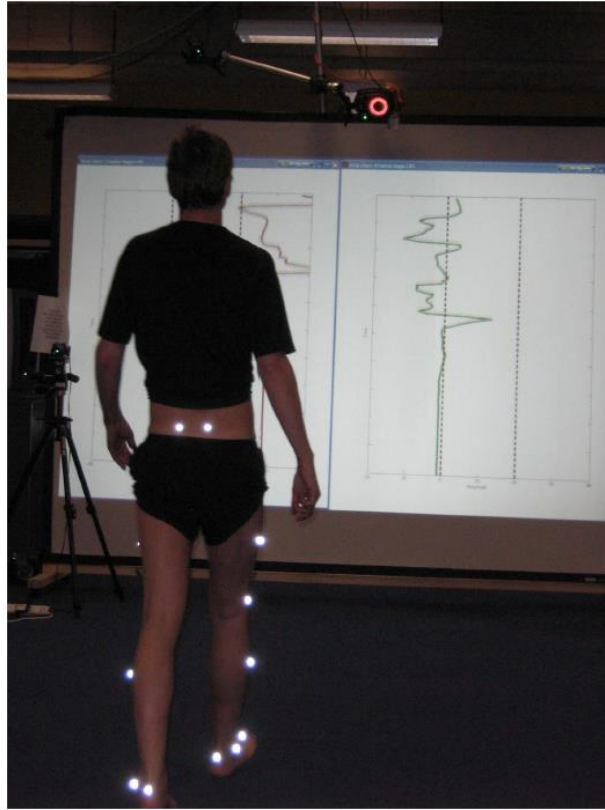


Figure 2.5 Movement biofeedback on toe out angle during walking gait

The work described here demonstrates that this is feasible and should be widely available. As a result, rather than requiring dedicated, often custom-built, and expensive biofeedback systems, movement biofeedback efficacy studies would be greatly facilitated by the availability of flexible biofeedback tools in conjunction with existing motion capture systems available in research labs and hospital gait analysis clinics. The method presented here reveals that this is entirely possible.

Another previous study (Pattamaset et al., 2020) used image-processing techniques to evaluate physical exercise for knee osteoarthritis. The researchers created an automatic system that processes images from knee exercise videos of the subject. This system utilized histogram analysis and skeletonization. The experimental result was tested by 17 subjects. This system verifies whether subjects do an exercise correctly or incorrectly. In the image acquisition section of this system, a standard camera records a subject in the sagittal plane while performing five knee exercises: active seated knee extension (A-SKE), active seated hamstring stretch (A-SHS), active heel slide knee extension (A-HSKE), active prone knee flexion (A-PKF), and actively assisted prone knee flexion (AAPKF).

In the image segmentation and classification section, the system converted the foreground image into a binary image and calculated the area of the white blob and the ratio of the white object's major and minor axis length to determine whether a current frame contains a human or not. The exercise posture classification begins by comparing the angle between the major and horizontal axes when a person is sitting or lying down. The exercise postures are then classified using template matching. They used histograms of binary image data to analyse human postures that estimated the position of the body, such as the knee, head, or foot, depending on each exercise posture.

After obtaining body position, they created a human skeleton by skeletonization with a thinning operation and then created a line to represent bone direction by using Hough transform with a subject's skeleton image to segment the line. In order to explain the subject's movement while exercising, the segmented line from the target position will be the estimated movement angle between the segmented line and either the horizontal axis or other, depending on an exercise posture. The feature extraction process for tracking exercise for A-SKE posture is summarized in Figure 2.6.

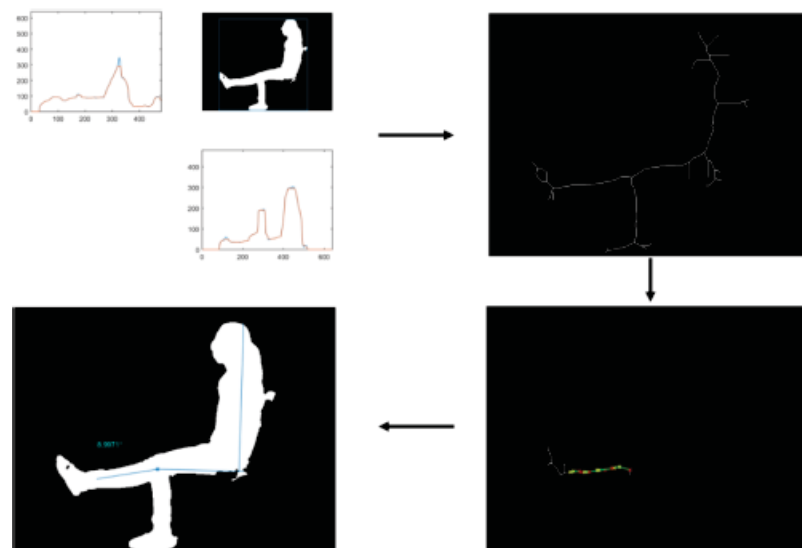


Figure 2.6 Feature extraction process of A-SKE posture

According to the authors, the overall accuracy of this automatic system is 88.12%. According to them, the error occurred in 11.88%, caused by the segmentation process, the classification process, the feature extraction and counting correct exercise function by 5.30%, 3.29%, 3.00%, and 0.29%, respectively. The background subtraction

technique caused the most errors (Pattamaset et al., 2020). Therefore, this research needs to improve the method of human subtraction for higher performance.

Furthermore, a previous study (Kang et al., 2019) proposed a Real-Time three-dimensional knee moment estimation method in Knee Osteoarthritis: toward biodynamic knee osteoarthritis evaluation and training. The authors of this study investigated differences in knee kinetic variables (external knee adduction, flexion, internal rotation moments, and impulses) during stepping on a custom elliptical trainer between patients with knee osteoarthritis (KOA) and healthy controls; and searched knee kinetic variables candidates for real-time biofeedback and complementing diagnosis/ evaluation on the elliptical trainer based on the knee injury and osteoarthritis outcome score (KOOS). Furthermore, they investigated the relationships between knee kinetic variables and 3-D ankle angles to investigate potential gait re-training strategies on the elliptical trainer. The knee kinetic variables and ankle angles were measured in real-time in a group of 10 patients with KOA and a control group of 10 healthy people of the same age and gender.

According to their observation, the elliptical trainer with the real-time biofeedback of EKAM (External knee adduction moment) (and other 3-D knee kinetic variables) can be used in clinics and gymnasiums with a few minutes of preparations and no post-processing of the motion analysis data. The results of this study warrant follow-up studies testing the efficacy of the KOA gait re-training on the elliptical trainer and the validity of the routine evaluation using the elliptical trainer. In the elliptical trainer, they used a 6-degrees-of-freedom goniometer underneath the footplate and according to their inspection, the 6-degrees-of-freedom goniometer can provide 3-D ankle angles at any sampling rate (Kang, Lee, Ren, et al., 2014), (Kang, Lee, & Zhang, 2014). It allows them to use the elliptical trainer outside motion analysis laboratories. The following Figure 2.7 represent the sagittal plane view of the elliptical trainer with the right leg.

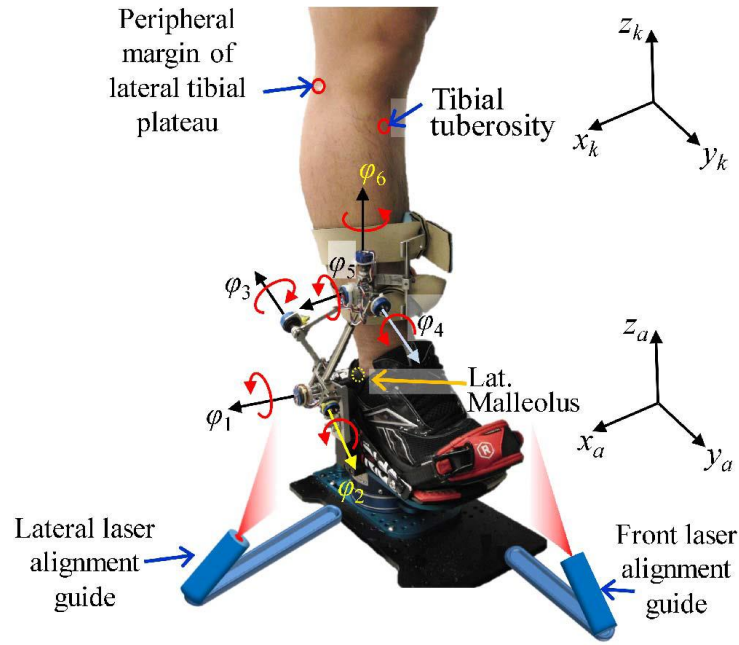


Figure 2.7 Sagittal plane view of the elliptical trainer

Finally, this system allows the authors to find real-time feedback of knee kinetic variables with potential strategies for modifying ankle angles, allowing them to immediately check the effectiveness of gait modification within a training session (Kang, Lee, Ren, et al., 2014), (Kang, Lee, & Zhang, 2014), and to achieve their long-term goal.

2.3.1 KSS and KOOS scores

The Knee Injury and Osteoarthritis Outcome Score (KOOS) and the Knee Society Score (KSS) are both clinical evaluation tools used to assess the functional status and quality of life of individuals with knee injuries or conditions such as osteoarthritis. The KOOS is a patient-reported outcome measure comprising five sub-scales, each assessing a different aspect of knee-related function: pain, other symptoms, function in daily living, function in sport and recreation, and knee-related quality of life. The scores for each sub-scale are combined to give a total score, with higher scores indicating better function.

The KOOS is intended to be used over short and long-time intervals; to assess changes from week to week induced by treatment (medication, surgery, physical therapy) or over the years due to a primary knee injury, posttraumatic OA or primary OA. KOOS, JR is scored by summing the raw response (range 0-28) and then converting it to an

interval score using the table below (Table 2.3). The interval score ranges from 0 to 100 where 0 represents total knee disability and 100 represents perfect knee health.

Raw summed score (0-28)	Interval score (0 to 100 scale)
0	100.000
1	91.975
2	84.600
3	79.914
4	76.332
5	73.342
6	70.704
7	68.284
8	65.994
9	63.776
10	61.583
11	59.381
12	57.140
13	54.840
14	52.465
15	50.012
16	47.487
17	44.905
18	42.281
19	39.625
20	36.931
21	34.174
22	31.307
23	28.251
24	24.875
25	20.941
26	15.939
27	8.291
28	0.000

Table 2.3 Table for converting raw summed scores to interval level scores from 0 (total knee disability) to 100 (perfect knee health)

		Baseline mean (SD)	Retest mean (SD)	Change scores mean (SD)	SEM	ICC agreement	95% CI
Mild OA (n = 35)	Pain	85.3 (18.5)	89.7 (12.5)	-4.4 (9.4)	7.2	0.80	0.60–0.90
	Symptoms	78.6 (7.1)	81.3 (16.8)	-2.2 (12.8)	9.0	0.74	0.54–0.86
	ADL	91.2 (15.1)	93.5 (10.8)	-2.3 (7.1)	5.2	0.85	0.71–0.92
	Sport/recreation	71.0 (23.4)	73.0 (22.9)	-1.7 (12.8)	9.0	0.85	0.73–0.92
	QoL	67.0 (21.6)	69.6 (20.5)	-2.5 (10.3)	7.4	0.88	0.77–0.94
Moderate OA (n = 53)	Pain	62.9 (25.7)	63.2 (23.7)	0.0 (12.9)	9.0	0.87	0.78–0.92
	Symptoms	63.2 (24.4)	66.2 (21.9)	-1.5 (11.3)	8.0	0.87	0.79–0.92
	ADL	69.3 (24.2)	69.2 (23.7)	0.6 (8.2)	5.8	0.94	0.90–0.97
	Sport/recreation	36.2 (32.0)	39.7 (32.5)	-4.8 (15.9)	11.6	0.87	0.78–0.92
	QoL	44.6 (26.3)	47.3 (25.4)	-2.1 (10.3)	7.4	0.91	0.86–0.95
Revision of TKA (n = 47)	Pain	61.6 (23.4)	63.9 (21.9)	-2.3 (14.3)	10.1	0.80	0.67–0.88
	Symptoms	64.7 (21.5)	62.9 (21.7)	1.7 (10.2)	7.2	0.89	0.81–0.94
	ADL	56.6 (21.9)	59.5 (22.9)	-2.9 (16.4)	11.7	0.73	0.56–0.83
	Sport/recreation	26.8 (34.1)	27.4 (34.2)	-2.2 (35.1)	24.6	0.45	0.19–0.66
	QoL	36.6 (26.6)	40.3 (27.5)	-3.7 (14.9)	10.8	0.84	0.73–0.91

Abbreviations: OA; osteoarthritis, SD, standard deviation; SEM, standard error of measurement; ICC agreement, intraclass correlation coefficient for agreement; CI, confidence interval. A normalized score (100 indicating no symptoms and 0 indicating extreme symptoms) is calculated for each subscale. Revision of TKA; Revision of total knee arthroplasty, AD; Activities of daily living; QoL; Quality of life.

Figure 2.8 Reliability of all subscales of the KOOS (Groot et al., 2008)

The above Figure 2.8 represents the reliability of all subscales of the KOOS score.

The KSS is a clinical evaluation tool used by healthcare professionals to assess the functional status of individuals with knee injuries or conditions. It consists of two parts: a subjective evaluation and an objective evaluation. The subjective evaluation includes a patient questionnaire, while the objective evaluation includes a physical examination of the knee. The scores for each part are combined to give a total score, with higher scores indicating better function.

2.4 The importance of guidance and assistance to improving quality, adherence and effectiveness of exercise.

One of the most important things people can do to improve their health is to engage in regular physical activity. Everyone, regardless of age, gender, race, ethnicity, or current fitness level, can benefit from moving more and sitting less. Physical activity is related to considerably more health benefits than we anticipated. Even better, small quantities of physical activity can have instant effects (U.S. Department of Health and Human Services, 2018).

After the physiotherapist has demonstrated an exercise to the patient, the patient typically does the exercise independently without further instruction or feedback. This study (Lam et al., 2014) suggests a method that provides patients with constant visual feedback and advice to improve their motion performance and adherence to

instructions. The system comprises body-worn inertial measurement units that measure the patient's position in real-time. During workout performance, the user sees the measured stance overlaid with the recommended movements on a visual display. Two user studies involving healthy volunteers were utilised to assess the usability of the visual guiding tool. The inertial measurement sensors were utilised to collect motion data and evaluate motion quality by comparing user performance with and without visual feedback and with or without workout guidance. According to the result of this study, there was a significant improvement in both quantitative and qualitative measures of exercise performance and technique when participants used the visual assistance tool.

Inexperienced exercisers perform core exercises without competent instruction, which has more destructive effects on them. Wearable sensors and algorithms are currently used in the medical domain to reduce that, but they must be more accurate to measure body segment kinematics and classify workout modes. A previous study (Lee et al., 2020) shows that deep learning improves squat posture classification, detects abnormal squat postures from Inertial Measurement Units (IMUs) data, and determines optimal IMU placement for self-fitness applications. This was achieved by the optimal placement of sensors at an appropriate location.

The authors tried to compare the squat posture classification performance of deep learning with a conventional machine learning algorithm. In order to do that, they used an accelerometer and gyroscope data from 39 healthy participants using five IMUs attached to the left thigh, right thigh, left calf, right calf, and lumbar region. According to the author's observation, concerning the convenience of use in self-fitness, the most feasible strategy was to utilize a single IMU on the right thigh. It was given an accuracy of 58.7% for conventional machine learning and 80.9% for deep learning. They said it is a better solution for self-coaching than wearing multiple IMUs.

Researchers have used some algorithms and learning techniques to practically use various coaching systems for Yoga. Recent studies in the literature have presented a yoga self-coaching system based on transfer learning techniques (Long et al., 2021). The yoga posture dataset was collected using a standard RGB webcam in the first step of this study, and then data augmentation techniques were used. The calculated angle of the joints achieved through keypoint estimation using the MediaPipe algorithm

(Long et al., 2021) is used to identify incorrect posture. The transfer learning technique was investigated and pre-trained on the MobileNet model. In the final stage, the authors built an AI yoga system based on a prediction model for real-time inference.

According to the authors, this yoga posture classification method achieved 98.43% accuracy by incorporating the yoga self-coaching system. This self-coaching system recognises yoga postures based on a selected guide to output predicted results and provide real-time guidance for incorrect postures. An artificial intelligence-based application might be useful to identify yoga poses and provide personalized feedback to help individuals improve their poses (Kothari, 2020). Another research (Goyal & Jain, 2021) proposed an algorithm to estimate the error in yogic poses by using the deep learning method. Simply said that they have proposed a method, using the Tensor-flow MoveNet Thunder model, that allows real-time pose estimation to detect the error in a person's pose, thereby allowing them to correct it. This study shows the model's results in real-time pose correction by training the Pose estimation algorithm on datasets of asanas to find the "perfect pose" and then proposing an algorithmic method to estimate the error in a yogic posture using angle calculations between joints.

Numerous studies on the impact of computer-assisted exercise on older persons' short- and long-term well-being have used commercially available technology or created bespoke systems to examine these effects. In a recent work (Doyle et al., 2010), the authors presented a proof-of-concept for an interactive vision-based system that provided strength and balance training to senior citizens. The authors only describe findings from limited laboratory testing, even though the proposed equipment was designed for senior usage in the home. Researchers have improved the precision of physical performance analysis by using appropriate devices and instruments.

Microsoft Kinect Camera is one such device that can be used to measure movement kinematics. It provides segmentation of human blobs even in cluttered scenes and real-time pose estimation with the 3D locations of 20 joints (Shotton et al., 2011). A previous study (Ofli et al., 2016) developed an automated interactive exercise coaching system using Microsoft Kinect for the well-being of the elderly. The coaching system walks users through a series of video exercises, monitors and records their every move, and gives them instantaneous feedback on how they are doing using the results of the human pose estimation taken from the Microsoft Kinect camera. This approach improves

balance, flexibility, strength, and endurance to reduce fall risk and improve daily performance.

The world's ageing population is increasing the demand for healthcare services. According to recent estimates, the number of persons over the age of 85 will increase more than fivefold by 2050 (US Census Bureau, 2022), and a nurse shortage is already becoming a concern (American Association of Colleges of Nursing (AACN), 2021)-(Buerhaus, 2008). As a result of this factor, recent designers and inventors have been able to incorporate human assistive robots successfully. One study (Fasola & Mataric, 2012) details the planning, execution, and assessment of a socially assistive robot (SAR) system meant to motivate older users to participate in physical activity to increase health and enhance their quality of life.

The researchers describe two user studies that were carried out to put the motivation theory to the test with their system. The first study compared a relational robot coach to a nonrelational robot coach to investigate the role of praise and relational discourse in the exercise system. The second study looked at participant preferences for user selection in the task scenario. Both studies assessed the robot exercise system's feasibility and overall effectiveness. In this system's configuration, the user is seated in a chair in front of the robot; the user and the robot face each other. The robot asks the user to perform simple seated arm gesture exercises during the exercise sessions. In the exercises, the robot's arm motion range is limited to the sides of the body to maximise the accuracy of the robot's visual detection of the user's arms. Figure 2.9 depicts the robot's platform and the exercise setup.



Figure 2.9 Robot platform and the exercise setup

To relate to this SAR system, the researchers primarily focused on seated aerobic exercise methods. They used a Microsoft Kinect camera to recognise real-time arm gestures/poses from the user. According to the researchers, the participants' experiences and feedback about the SAR exercise system were highly positive.

Another study (Ono et al., 2014) proposed a motion modelling method for evaluating motion patterns during exercise. The authors used a steady-state genetic algorithm to estimate the arm joint angle of the robot partner, which corresponded to human motion, and a neural network to model the motion pattern. The authors presented some experimental results in the results and analysis that discussed the proposed method's effectiveness in evaluating a range of motion in an exercise programme with a robot partner. They have already decided to implement a robot conversation system to increase older people's sense of self-efficacy in future work.

2.5 Study review of human body posture and movement tracking devices and technologies.

The dynamic detection of human motion is critical and widely used in motion state capture and rehabilitation engineering. A human body posture detection system detects changes in posture in a specific area and the range of human motion. Generally, it can be quantified by the devices or by employing a specific technology. Human posture detection is classified into two types based on different detection approaches: nonwearable detection and wearable detection (Li et al., 2021). Previous studies (Li et al., 2021) on wearable detection technology proposed multimodal information of surface electromyography (sEMG) signals of the upper limb, triaxial acceleration, and plantar pressure signals of the lower limb as the most efficient virtual driving control and gait recognition techniques. Showing the effectiveness of wearable human posture detection has also been a part of this goal. According to this study, the upper limb sEMG signal analysis was used in virtual driving. The virtual driving operation was carried out using effective feature extraction and classification identification based on the real-time and safety characteristics of the sEMG signal (Farina et al., 2014). An optimised support vector machine (SVM) classifier was used to obtain effective gait identification results by combining the features of triaxial acceleration and plantar pressure.

Another study (Chen et al., 2021) proposed a system for at-home physical therapy that classifies single-axis spinal motion using a wearable system of stretch sensors. The

researchers successfully implemented that system by enabling real-time biofeedback to ensure the proper execution of physical therapy exercises at home and the lightweight stretch sensors used to estimate a patient's spinal posture. The sensor data was used to train and test an SVM-based (support vector machine) classification algorithm. According to the results and analysis, the average accuracy of this proposed system was up to 80%.

Furthermore, researchers (Adochiei et al., 2019) developed and validated a posture detection system based on two sensor modules and positioning sensors. The implemented device can deduce information about the user's changing posture over time by collecting sensor readings in real-time and then applying a series of algorithms to filter and process the data. This study also aimed to obtain additional positioning functions, like checking body posture and the self-correction posture for reducing the wrong posture automatically, to strengthen the back muscles and obtain health benefits. These benefits, and implicitly, the improvement of the health status, can be obtained through body posture monitoring devices. The final product is a portable, non-invasive device, has a small size and relatively low weight, is suitable for attachment to the human body, and is characterised by high accuracy.

2.5.1 Non-wearable sensors

Researchers were inspired to focus on non-wearable devices despite the reasonable accuracy of the wearable human posture detection system due to the challenge of keeping them planted (Guo et al., 2017)-(Bian et al., 2019). Furthermore, some studies have found that adherence to wearable devices decreases over time (Lazar et al., 2015). For these reasons, researchers were interested in tracking human postures and movements in three-dimensional space. The researchers could do that quickly thanks to the invention of depth cameras, which provide clear information on 3D structural views. Human skeleton joint data is provided by cameras with depth sensors such as Microsoft Kinect, RealSense, and Orbbec Astra pro (Farooq & Won, 2015). The research community is currently focused on using skeleton joint data for human posture and motion analysis. Figure 2.10 depicts the Microsoft Kinect and the Orbbec Astra pro-depth-sensing cameras.



Figure 2.10 3D depth sensing cameras

A study (Tolentino et al., 2019) used the Microsoft Kinect sensor to create a system that recognises various emergencies based on body gestures. The system was able to recognise 14 different gestures that can be associated with various emergencies. After detecting the gestures, the system recognised the type of gestures performed by the victim using the decision tree created by the researchers. The decision tree's conditions caused this. Angles and distances were computed using mathematical representations. According to the authors' findings, the mathematical representations were sufficient for the sensor to determine what type of gesture was performed. According to the analysis of the results, some gestures have low reliability because a slight variation in the movement shifted the recognition from one gesture to another, whereas standing gestures have high reliability. After all, the camera can easily find all joints required (Tolentino et al., 2019).

With the discontinuation of the Microsoft Kinect series, the Orbbec Astra has become the most common depth-sensing device in human motion analysis due to its availability and gesture recognition software, which can be quickly developed using the provided Software development Kit (Coroiu & Coroiu, 2018). However, it is currently quite expensive, so its use in research is minimal. For these reasons, researchers showed interest in using non-wearable devices, sensors, and other technologies.

Guanlong Teng et al. (Teng et al., 2020) propose MobiFit, a freehand exercise assistant system that does not rely on any wearable device or sensor but instead on a single cellular signal receiver. This system's goal is to monitor freehand exercises using ubiquitous cellular signals and to provide exercise quality indicators such as repetition number, exercise type, and duration for each exercise repetition. It operates in a

contactless mode, counting and recognising six freehand exercises and providing exercise repetition and rest interval distribution for each session to assess workout quality. MobiFit processes the down-sampled cellular signals to count and recognise exercise repetitions. MobiFit's framework is depicted in Figure 2.11, which includes five modules: signal receiving, segmentation, feature extraction, classification, and output.

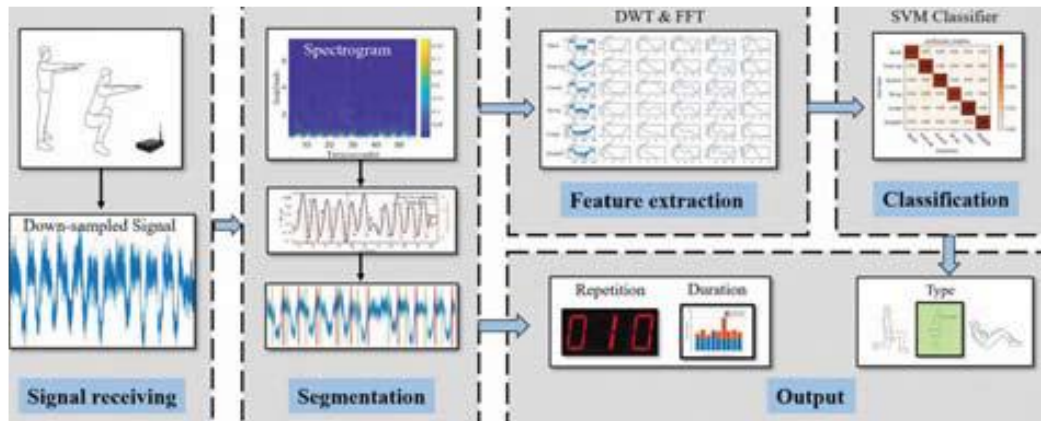


Figure 2.11 Framework of the MobiFit

Furthermore, the experiments confirm that MobiFit is feasible for use indoors and outdoors, as well as its ability to support multiple users exercising simultaneously.

Another study (Konstantinidis et al., 2016) implemented an exergaming platform for senior users called 'FitForAll'. This platform is composed of aerobic, strength, balance and flexibility computerized exercises blended with games. Each game is a goal-oriented activity through a virtual environment with simple and understandable graphics. The Microsoft XNA Game Studio (Nandi, 2022) was employed for the rendering of 2D and 3D graphics of this application. The platform consists of four difficulty levels the first level promotes light exercise, while the last promotes more intense physical exercise. Overall, this programme has allowed some exercise to be done enjoyably; however, due to the limitations they have noted, comparing this to traditional physical activities is tough. Besides this, recognizing human posture and movement through machine learning and computer vision is now improving rapidly. Here, human skeleton points detection algorithms make it possible to perform body posture analysis and movement tracking with a high degree of accuracy and precision.

A human skeleton points detection algorithm is a computer program that can detect key points on a human skeleton in an image or video. These points typically include joints such as the wrists, elbows, shoulders, hips, knees, and ankles. The purpose of such an algorithm is to enable the analysis of human movement and posture, which has applications in fields such as biomechanics and human-computer interaction. Although human skeleton point detection frameworks are common, they have some differences in terms of their features and capabilities (Carey et al., 2020). MediaPipe is an open-source framework that focuses on real-time computer vision and machine learning (Boesch, 2022b). It has a modular design that allows developers to easily add new modules and customize the pipeline according to their needs. MediaPipe has pre-built modules for skeleton point detection and can accurately detect the locations of key points on a person's body in real-time.

A recently published paper (Ma et al., 2022) proposes a Wushu posture recognition system for tracking user hand, head, and body movements using a camera and MediaPipe. The authors created recognition algorithms for the Fist, Palm, Hook, Tiger Talon, Forward Lunge, Horse Stance, and Empty Step, according to the Landmarks returned by MediaPipe. They used Python to create a system that allows users to perform Wushu training independently, safely, and efficiently without the assistance of a teacher. According to their observations, the overall experimental results of 400 photos show that these algorithms can effectively identify the movements mentioned above.

TensorFlow is also an open-source framework developed by Google that is widely used for a variety of tasks, including computer vision and natural language processing. It has a large community of users and developers, making it easy to find resources and support for working with TensorFlow. TensorFlow can be used for skeleton point detection using its MoveNet models (TensorFlow.js, 2021), but it may require more development work and customization compared to MediaPipe. Using TensorFlow MoveNet, a previous study (Joshi & Joshi, 2021) proposed a predictive neural network (NN) for joint angle profile prediction across variable walking speeds and slopes.

The proposed model successfully predicted knee joint angle profiles at three different walking speeds (0.8, 1, and 1.2 m/s) and nine AOI (Angle of Inclination) values ranging from -10 to 10. MoveNet achieved an RMSE (Root Mean Squared Error) of 3.24, 1.19

and an MAE (Mean Absolute Error) of 2.66, 1.00 in the results of this system, based on ten-fold cross-validation. These findings imply that artificial intelligence can predict subject-specific knee joint angle profiles for variable slopes at a given walking speed based on data from the flat-surface knee joint angle profiles.

Aside from these two frameworks, OpenPose is a framework that focuses on human pose estimation. It employs deep learning algorithms to precisely locate key points on a person's body, including skeleton points. OpenPose is designed for real-time performance and can be used in motion capture and video surveillance applications (Boesch, 2022a). A previous study (Chen & Yang, 2018) introduces Pose Trainer, an end-to-end computer vision application that uses pose estimation, visual geometry, and machine learning to provide personalised feedback on fitness exercise forms using the OpenPose framework. The authors evaluate exercise videos through human pose key points using OpenPose's output. Pose Trainer uses state-of-the-art pose estimation to detect a user's pose and then evaluates the vector geometry to provide feedback. Based on personal training guidelines, they record over 100 exercise videos of correct and incorrect forms to evaluate.

2.5.2 Transfer learning technology

Transfer learning technology is also used for image classification. It can improve the performance of a skeleton point detection algorithm by fine-tuning a model previously trained on a similar task, such as human pose estimation. Based on transfer learning, a study (Ke et al., 2021) proposed a visual analysis system of human morphological action behaviour detection in fitness exercises. Based on multi-scale feature decomposition, the proposed method detects human action behaviour characteristics in fitness exercises. Using multimedia vision, they primarily collected images and performed feature analysis on complex action links in fitness exercises.

Harris feature point detection and information screening methods were then applied to improve the image of human action in fitness exercises. Multi-scale morphological edge contour features of the image were extracted to calibrate and detect the feature points of the motion behavior features of the body-building exercise and improve the accuracy of quantitative estimation of motion behavior. The feature points of human motion behavior characteristics of this system as shown in the following Figure 2.12.

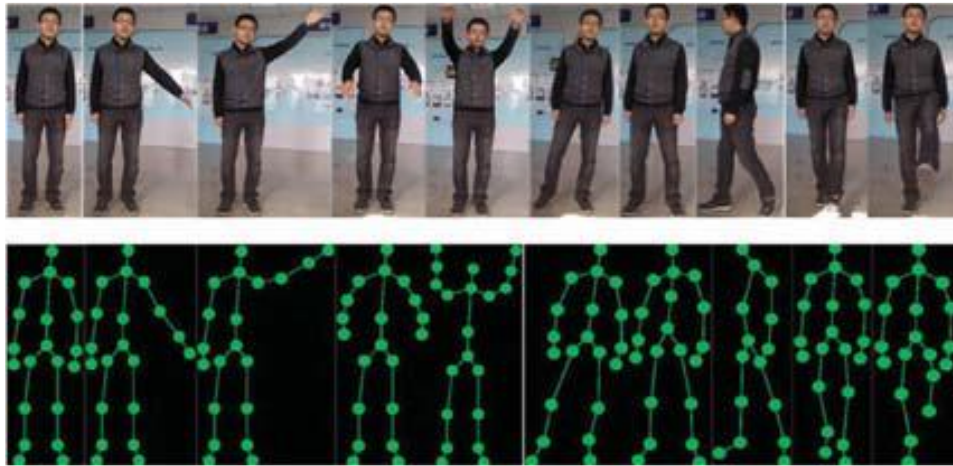


Figure 2.12 Feature points of human motion behaviour characteristics

According to their findings, this simulation demonstrates that this method has a high detection probability, increases the number of crucial information feature points, and has a high application value in the correction of fitness training action norms.

2.6 Body posture detection system used in sports and medical domains.

Various body posture detection systems are used in sports and the medical domain. Sensors and cameras are typically used in these systems to track the movement and positioning of a person's body in real-time, allowing the detection of any abnormalities or potential injuries (Wiedemann et al., 2015). These systems are frequently used in sports to help athletes improve their performance and avoid injuries, while in medicine, they are used to help diagnose and treat a variety of conditions. Motion capture systems, biomechanical analysis systems, and posture analysis software are examples of body posture detection systems.

Biomechanical analysis is the study and science of how biological creatures' structures respond to external forces and stimuli (Lu & Chang, 2012). According to a previous study, biomechanics in sports is done by analysing the physiological mechanics of athlete movement (Landry, 2014). Many researchers in the literature have identified practical applications of biomechanics in athlete movements with the following primary goals.

- Improve athletic performance by identifying and applying the optimal technique
- Prevent injury and speed up recovery
- Identify muscle imbalances

Athletes can move by running, jumping, walking, and so on. A movement usually takes less than a few seconds to complete. According to the literature, identifying a movement in a set of frames is a concern. Several studies have been proposed to visualise and quantify athlete movements as computer vision technology advances.

Pueo et al. proposed a method (Pueo, 2016) that uses high-tech cameras to visualise an athlete in 3D space. The frame-by-frame analysis is used to determine the angles and positions that an athlete moves in. This analysis is carried out in an indoor setting, at a specialised laboratory. Rapid movements such as football kicks and indoor cycling are considered in this study. The literature suggests that if the motion analysis lab possesses a force plate or a force treadmill, it is possible to calculate the ground reaction force of athletes when they perform movements.

Keng et al. (Chang et al., 2007) used a 3-axis accelerometer embedded in a workout glove to detect weight-lifting athletes' hand movement. In this study, the athlete's free weight exercises are identified, and the number of repetitions is counted. The accelerometer's output is streamed and fed into a type recogniser. The streamed data represent acceleration along the x, y, and z axes. The Naive Bayes classifier and Hidden Markov models are used to classify the exercise based on the highest likelihood. The accuracy of the exercise is not taken into account in this approach. The work limits only to recognize the exercise based on the highest likelihood.

The study (Lu et al., 2021) proposes a method to recognize the human lower limb vertical jump phases using bidirectional LSTM (Long Short-Term Memory System) and convolutional LSTM. This work limits the recognition of the vertical jump phase based on previously trained jump phases.

A previous study (Rahman et al., 2013) proposed a Multimedia Interactive Therapy Environment for Children Having Physical Disabilities. The study aimed to learn and correct a physiotherapy exercise without physical inspection. The users can mimic the character's movement in their Kinect-based online second life to work on physical therapy. The study focuses only on elbow joint variation and compares the angle calculation with a conventional goniometer device.

Another study (Ding et al., 2017) developed 'FEMO', A Platform for Free-weight Exercise Monitoring using RFIDs, which determines what exercise a user is

undertaking based on the frequency of the weight's movements. They discovered that each workout has its frequency, allowing them to identify the exercise by looking at the wavelengths. Even though the proposed work identifies the free weight exercise, this work fails to identify the correctness of the kinematics with time.

Body posture detection systems have not been widely used in the medical domain, but researchers have proposed some recommended systems for medical-related exercises. Research (Patil et al., 2022) introduced a free weight exercise tracking software that allows users to learn and correct their form. The squat exercises such as squats, push-ups and bicep curls were chosen for this study because they have particular needs that could be tracked and modified easily. The researchers have already stated in their future works that it is necessary to consult a physician or physiotherapist who can verify whether the exercise is proper and that some exercises, such as push-ups, had lower accuracy owing to the camera quality used for the application.

A study (Lai et al., 2009) proposed a system of Automatic Recognition of Gait Patterns Exhibiting Patellofemoral Pain Syndrome Using a Support Vector Machine Approach. A CI approach (SVM) was applied to recognize PFPS gait patterns from a large set of gait features extracted from GRFs (Ground Reaction Force) and rearfoot and tibia kinematics. According to their observation, the overall test results indicated that the SVM could effectively recognize gait patterns that were affected due to the PFPS condition from a small combination of gait features with 89% accuracy. The study was designed only for female gait pattern recognition.

Another previous study (Chiang et al., 2018) shows an exercise guidance system for "The Optimal Lymph Flow" that automatically detects whether a user performs a set of exercises correctly based on depth and joint positions captured by a Kinect sensor. A Kinect sensor and a MOCAP system are used to train a Gaussian Process regression model for denoising a Kinect joint captured by a Kinect sensor into a standardised domain. This work has completed the algorithm design for four exercises related to lymphatic health and lymphedema intervention. However, the fact that Microsoft has currently discontinued Kinect cameras has a significant impact on this.

Aside from these systems, a study (Thiengwittayaporn et al., 2021) proposed the "Rak Kao" (English translation: Love-Your-Knee) mobile application to analyse

questionnaire data and generate a personalised recommendation of treatment and exercise type using rule-based and Artificial Intelligence (AI) techniques. This proposed work assessed accuracy in the performance of three prescribed knee exercises such as catch-bend-down, stretch-touch-feet, and sit-stretch-hold. Clinical outcomes were assessed using the KOOS and KSS scores before and after the 4-week programme to assess the range of motion, symptoms, pain, physical activity, and quality of life. As they have stated, the patients in this study were able to use smartphones efficiently, which may not be representative of the larger elderly population.

2.7 Chapter Summary

The literature review covers five essential features of the current study. First, the impact of exercises on the recovery of osteoarthritis patients was investigated, and it was discussed in further depth how exercises might help improve symptoms, overall function, and quality of life. Here, the research conducted in accordance with international guidelines such as ACR, EULAR, and OARSI has been described in depth and the effects of exercise on osteoarthritis patients have been shown through their conclusions. The next section consists of previous research studies on the implemented devices and technologies used for osteoarthritis. Here mainly, the wearable and non-wearable devices and technologies presented by the researchers, their obtained results and analysis, and some limitations have been discussed.

Recognising the patient's posture using a three-dimensional space rather than the two-dimensional image classification and feature extraction method in motion analysis can be done with higher accuracy. However, it requires some special wearable sensor systems or 3D capturing cameras that are very expensive on the market. To correctly perform physical exercise, one should follow the instructions of a well-trained guide. The third section clarifies it further with relevant research. Researchers proposed more accurate physical and virtual exercise guidance and systems with positive participant feedback to achieve the same goal.

Wearable and non-wearable detection are primarily classified in the introduction to the devices and technologies used for human body postures and movement tracking. Aside from these, researchers are currently focusing on non-wearable human body postures, movement-tracking devices, and technologies. The fourth section goes into great detail about how the researchers made the most effective efforts. With the advancement of

technology, tracking human body movements using Artificial Intelligence is developing rapidly. Human skeleton point detection is a process of identifying and locating specific points or joints on the human skeleton that can be done using computer vision algorithms and machine learning techniques. It can analyse the visual data and detect the locations of the various bones and joints in the body. Many improved frameworks are currently available for this purpose, and researchers testify that the framework known as 'MediaPipe' has a very high level of accuracy.

The final section discusses how body posture detection systems have been used in sports and medicine. The medical domain has very few human skeleton point detection systems that use computer vision and ML algorithms. Under standard medical and clinical recommendations, no existing model or device has been implemented to measure and analyse human body angles and point-to-point distance between specific key points on the human body. Previous research studies have not found a more cost-effective solution for guiding patients with knee osteoarthritis by properly measuring their knee joint angles and guiding them to perform the exercise. Given these facts, this study aims to reduce the disparity by developing a solution that can be used to self-evaluate the knee joint angles according to the relevant exercise and guide the patient to achieve the daily target properly.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter outlines the methodology employed to conduct the research in alignment with the aims and objectives as delineated in Section 1.4. The overarching goal of this study is to develop a Virtual Physiotherapy Guider capable of autonomously evaluating the accuracy of rehabilitation exercises performed by individuals with knee osteoarthritis. Research questions have been formulated to steer the inquiry into the effectiveness and feasibility of the proposed system. The chapter elucidates the various methods employed to gather and analyze data, aiming to provide solutions to each of the research questions. The envisioned framework integrates biomechanical assessment and performance parameter evaluation. The high-level architecture of this study illustrates a stepwise procedure leading to the study's conclusive findings. Figure 3.1 visually represents the high-level architecture of this research.

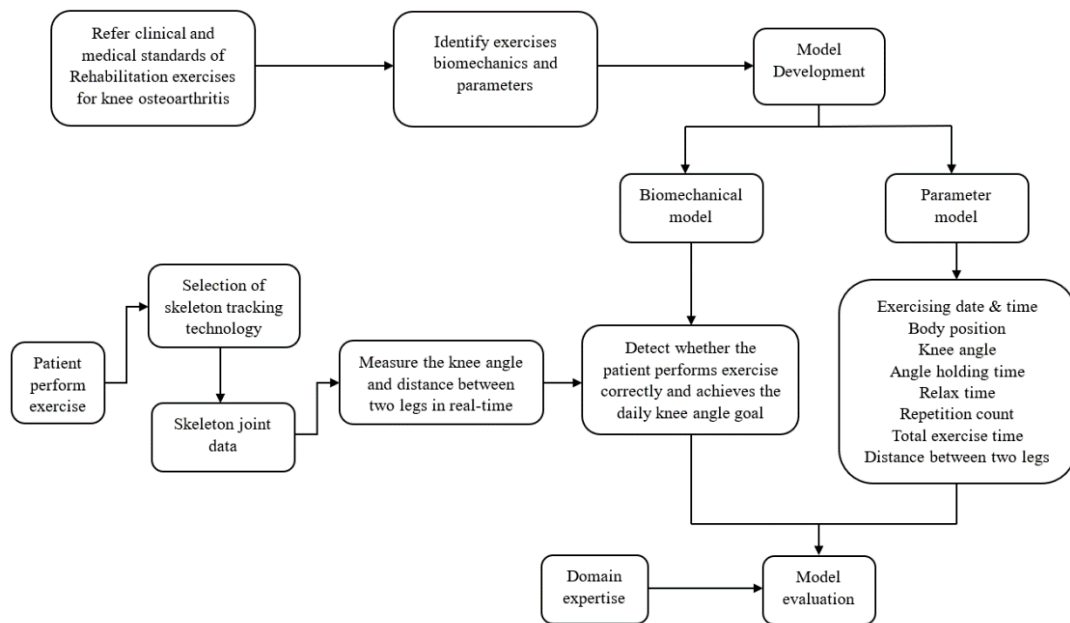


Figure 3.1 High Level architecture of the research design

According to the Norkin and Joyce (2016), the knee joint's range of motion (ROM) stands as a crucial clinical parameter extensively utilized in knee assessments. Despite its significance, existing literature and online sources lack a comprehensive and

quantified description of a model that seamlessly integrates both the detection of body position during exercises and the provision of real-time feedback. While conventional methods such as goniometry contribute to understanding knee flexion, there exists a gap in the integration of modern technologies to create a holistic system capable of evaluating exercises in real-time. This research seeks to bridge this gap by proposing a Virtual Physiotherapy Guider. The guider not only measures the knee joint's range of motion but also incorporates advanced technologies for accurate body position detection during exercises. This integration offers invaluable real-time feedback to individuals with knee osteoarthritis engaged in rehabilitation activities, achieved through the translation of theoretical knowledge from various sources into practical and actionable mathematical conditions.

In clinical settings, Goniometers have long been a staple for assessing the knee joint's range of motion in patients with knee osteoarthritis. These manual instruments provide a direct and measurable means of quantifying the flexion and extension angles of the knee. A healthcare professional typically places the Goniometer on specific anatomical landmarks, such as the hip, knee, and ankle joints, and then reads the corresponding angle to evaluate the patient's knee movement.

While Goniometers have demonstrated reliability, recent technological advancements have ushered in alternative methods for measuring and monitoring knee joint movement and angle. Wearable sensors and inertial measurement units, as revealed by an extensive literature review conducted in the preceding chapter, have become key components in this domain. Researchers have successfully developed both wearable sensing units and non-wearable movement detection systems. These innovations enable continuous monitoring of knee osteoarthritis patients' movements during various activities, eliminating the constraints associated with traditional measurement tools. Through the integration of modern technology, these systems offer a more versatile and comprehensive approach to assessing knee joint dynamics, providing valuable insights for both clinicians and patients alike.

However, existing literature and online sources lack a quantified description of the ideal biomechanics required for measuring the knee angle without the use of any sensor unit. Therefore, this research has taken on the task of developing its own description of the biomechanics involved by translating theoretical knowledge from various sources into

mathematical conditions. To assess the biomechanical accuracy of patients' exercise movements, the research will employ a mathematical approach. This involves computing the distances between points that form lines representing different body parts and calculating the angles between these lines. For instance, when measuring the knee angle, careful consideration must be given to the positions and distances of the hip, knee, and ankle joints. The knee joint angle will be continuously monitored against specific mathematical conditions using the performance parameter model. If the calculated angles meet the clinically accepted defined conditions, the exercise can be deemed to meet the required standards. This quantitative evaluation approach offers a precise method for assessing the biomechanics of exercise.

In addition to monitoring the knee angle, the performance parameter model captures data such as the date and time of the patient's exercise, the patient's body position, the duration of holding the knee angle during the exercise, relaxation time, the number of repetitions, total exercise duration, and the distance between the legs. This model comprehensively gathers information to provide a detailed analysis of the exercise performance. In connection with the 'Biomechanical Model,' the knee angle derived from the performance parameter model plays a crucial role in monitoring the correctness of the patient's exercise. It serves as a key metric to assess whether the patient is achieving the daily knee angle goal associated with that specific exercise.

Several Human Pose Estimation technologies are available to identify skeletal points from a given image or video input. The selection of an appropriate technology to detect the necessary skeletal points from a patient's body is crucial, particularly in identifying angles, especially on the knee joint. This capability allows for the analysis of whether the exercise is performed correctly, utilizing both the biomechanical and performance parameter models. Therefore, choosing the right technology is essential to obtain accurate and reliable data for further analysis. Before selecting a skeleton tracking technology, it is crucial to carefully consider several key factors. These considerations play a pivotal role in ensuring the effectiveness and suitability of the technology for the specific requirements of the research. Here are the significant points that need to be considered before selecting a suitable skeleton tracking technology for measuring the angle of the knee joint.

3.1.1 The significant points when choosing a skeleton tracking technology to measure the Knee Angle

There are several important considerations to keep in mind when choosing a skeleton tracking technology to measure the angle of the knee joint:

- **Accuracy:** It is important to select a technology that can accurately track the knee joint's movement and position, as small tracking errors can lead to significant errors in the measured angle.
- **Range of motion:** The technology should accurately track the full range of motion of the knee joint, including flexion, extension, and any rotational movement.
- **Sensitivity:** The technology should be sensitive enough to detect small changes in the angle of the knee joint, as these can be important indicators of injury or other problems.
- **Ease of use:** The technology should be easy for the patient and the healthcare provider or researcher collecting the data.
- **Cost:** The technology should be cost-effective, particularly if used on many patients or in a long-term study.
- **Compatibility:** The technology should be compatible with other equipment or systems that will be used in conjunction with it, such as motion capture systems or electronic health records.
- **Reliability:** The technology should be reliable, with minimal downtime or maintenance required.

In selecting an optimal technology for our research, we have carefully assessed various Skeleton Tracking Technologies, choosing one that stands out prominently among available options. This technology serves as a robust foundation for developing a rule-based approach to create both the biomechanical model and the performance parameter model. The rationale behind adopting a rule-based approach is to encapsulate the expertise found in human experts and medical journals, specifically within the medical and healthcare domain. This approach enables the encoding of knowledge into computational rules.

In addition to technological considerations, the research undertakes the crucial task of identifying clinically accepted rehabilitation exercises specifically tailored for

individuals with knee osteoarthritis. This aspect of the study involves a thorough exploration of established medical literature, expert recommendations, and clinical guidelines within the domain of knee osteoarthritis rehabilitation.

All patient data acquired from these developed models will be systematically saved in a CSV file using the patient's name for comprehensive monitoring. Through these methodologies, we anticipate the creation of a Virtual Physiotherapy Guider for Knee Osteoarthritis using Computer Vision and Machine Learning. This innovation is expected to play a pivotal role in encouraging individuals with knee osteoarthritis to become more physically active.

3.1.2 Identify clinically accepted rehabilitation exercises for knee osteoarthritis

Identifying clinically accepted rehabilitation exercises for knee osteoarthritis is a critical aspect of tailoring effective intervention strategies. This process involves an exhaustive review of reputable medical literature, evidence-based guidelines, and expert consensus within the field of knee osteoarthritis rehabilitation. By drawing on the collective knowledge of healthcare professionals and established protocols, the goal is to curate a set of exercises that have demonstrated safety, efficacy, and positive outcomes for individuals with knee osteoarthritis. Clinically accepted rehabilitation exercises for OA of the knee may include followings:

- **Quadriceps strengthening exercises:** These exercises target the muscles in the front of the thigh (quadriceps), which are important for knee stability and function. Examples of quadriceps strengthening exercises include leg presses, leg extensions, and squats.
- **Hamstring strengthening exercises:** The muscles in the back of the thigh (hamstrings) also play a role in knee stability and function. Hamstring strengthening exercises can include leg curls and glute bridges.
- **Low-impact aerobic exercises:** Aerobic exercise can help improve cardiovascular fitness and promote weight loss, which can reduce the burden on the knee joint. Low-impact options include swimming, biking, and walking on a treadmill or flat surface.
- **Range of motion exercises:** These exercises can help improve flexibility and joint mobility. Examples include leg lifts, leg swings, and calf stretches.

- **Balance and coordination exercises:** These exercises can help improve balance and stability, which can help prevent falls and injuries. Examples include single leg stands and Tai Chi.

Among the five main exercise types, the literature review conducted for this study has identified eight specific exercises. These exercises are categorized into three primary types: Stretching, Strengthening, and Flexibility. Furthermore, in alignment with the physical therapy plan, each of these exercise types is stratified into different levels, denoted as Level 1, Level 2, and Level 3. These levels correspond to beginner, intermediate, and advanced stages, respectively. Below, we outline the selected exercises for each of these levels, ensuring a comprehensive approach to rehabilitation that caters to individuals at different proficiency levels.

Level	Level Description	Selected exercise
Level 1 (Beginner)	Therapeutic strengthening for beginners.	• Inner Range Quadriceps. • Knee extension (full arc quadricep strengthening).
Level 2 (Intermediate)	Knee Strengthening Exercises.	• Straight leg raises. • Standing hamstring curls.
Level 3 (Advanced)	Dynamic Strengthening Exercises.	• Reverse lungs.

Table 3.1 Selected exercises related to the levels of the physical therapy plan.

The selected exercises, corresponding to the defined levels (Level 1 for beginners, Level 2 for intermediate, and Level 3 for advanced), are tailored to the specific stage of the patient. Recognizing the varying capabilities and needs of individuals at different stages of their rehabilitation journey, these exercises are strategically chosen to address the gradual progression of difficulty. This approach ensures that patients are provided with exercises that align with their current proficiency level, facilitating a customized and effective rehabilitation program. By adapting the exercises to the patient's stage, the rehabilitation process becomes more personalized, promoting optimal engagement and adherence to the prescribed regimen.

3.1.3 Selecting the suitable skeleton tracking technology

Taking into account the existing literature on technologies, we conducted an independent study by reviewing the official documentation of several Human Pose Detection libraries. The results of this investigation are presented in Table 3.2.

Technology	Single/ Multiple	Dimension	Approach	CPU/ GPU	Flickering/ Jitter	Body Key Points	Speed	Optimized
BlazePose	Single	2D, 3D	Bottom-up	CPU	Low	33	High	Yes
OpenPose 2D	Multiple	2D	Bottom-up	GPU	Low	135	Low	No
OpenPose 3D	Single	3D	Bottom-up	GPU	Low	135	Low	No
AlphaPose	Multiple	2D, 3D	Top-Down	GPU	High	26	Low	No
YOLOv7	Multiple	2D, 3D	Top-Down Bottom-up	GPU	High	17	Low	No

Table 3.2 Comparison Matrix

BlazePose (Bazarevsky et al., 2020), OpenPose 2D (Cao et al., 2018), OpenPose 3D (Nakano et al., 2020), AlphaPose (Fang et al., 2022), and YOLOv7 (Wang et al., 2022) stand out as some of the most popular computer vision technologies employed for human pose estimation. In this section, we will conduct a comparative analysis of these technologies, considering various metrics such as single or multiple-person detection, 2D and 3D perspective, approach, CPU/GPU usage, flickering, key points, speed, and optimization for CPU-edged devices.

- **Single or Multiple person detection:** BlazePose is optimized for both single and multiple-person detection, making it a versatile choice. OpenPose 3D excels in single-person detection, while OpenPose 2D, AlphaPose, and YOLOv7 are tailored for multiple-person detection scenarios.
- **Perspective:** In terms of perspective, OpenPose 2D focuses on 2D, while OpenPose 3D, BlazePose, AlphaPose, and YOLOv7 are optimized for both 2D and 3D perspectives.
- **Approach:** Regarding the approach, BlazePose, OpenPose 2D, and OpenPose 3D utilize a bottom-up strategy by detecting individual key points and assembling them into a complete body pose. In contrast, AlphaPose and YOLOv7 employ a hybrid top-down and bottom-up approach, first detecting the person and then key points.

- **CPU/GPU Usage:** For CPU/GPU usage, BlazePose is CPU-optimized, while OpenPose 2D, OpenPose 3D, AlphaPose, and YOLOv7 excel in GPU usage. BlazePose uniquely supports CPU-edged devices, providing accessibility without requiring a GPU.
- **Flickering:** When it comes to flickering, BlazePose, OpenPose 2D, and OpenPose 3D offer low flickering, ensuring a smooth video output. In contrast, AlphaPose and YOLOv7 may exhibit higher flickering, impacting video output quality.
- **Body Key Points:** Key points availability varies with BlazePose offering 33, OpenPose 2D and 3D providing 135 each, AlphaPose offering 26, and YOLOv7 providing 17 key points.
- **Speed:** In terms of speed, BlazePose stands out as the fastest technology, offering high processing speed. OpenPose 2D, OpenPose 3D, AlphaPose, and YOLOv7 are comparatively slower.
- **Optimized for CPU Edged Devices:** BlazePose is the only technology that is optimized for CPU-edged devices, which means that it can run on devices such as smartphones and laptops without a GPU. The other technologies require GPUs for optimal performance and cannot be run on CPU-edged devices.

Overall, BlazePose seems to be the best technology for human pose estimation in terms of single or multiple-person detection, low flickering, optimised for CPU-edged devices, and high processing speed. However, it's important to note that when considering 3D perspective, the use of specialized 3D depth sensing cameras is mandatory, which may introduce higher costs. Currently, due to budget constraints, these 3D cameras are not commonly used in low-cost research endeavors.

3.1.4 Selection of the MediaPipe Framework for Skeletal Point Extraction

After conducting an exhaustive literature review for this research, we have opted to proceed with MediaPipe HPE for several compelling reasons. MediaPipe boasts numerous advantages over alternative skeletal estimation technologies, including OpenPose, PoseNet, and MoveNet. Here are some of the reasons why MediaPipe stands out as a preferred choice:

- **Accuracy:** MediaPipe is known for its high accuracy in detecting and tracking human body joints and movements. It is based on machine learning algorithms that make it highly accurate in detecting even the slightest of movements.
- **Cross-platform support:** MediaPipe is a cross-platform technology that supports a wide range of devices, including smartphones, laptops, and desktops. It also supports different operating systems like Android, iOS, and Windows, making it a versatile option for developers.
- **Easy to use:** MediaPipe comes with a simple API that makes it easy for developers to integrate it into their applications. It also provides pre-trained models that can be easily used without any prior knowledge of machine learning.
- **High-speed processing:** MediaPipe uses Tensor Processing Units (TPUs) for high-speed processing, which enables it to perform the real-time skeletal estimation.
- **Customizability:** MediaPipe provides some level of customizability, allowing developers to modify the models to fit their specific needs. Although this may not be as extensive as some other skeletal estimation technologies, it still provides some flexibility.
- **More landmarks:** MediaPipe consist of 33 landmarks when compared to other libraries such as Movenet and PoseNet.

Numerous studies have utilized the MediaPipe platform for developing computer vision and machine learning models, revealing its superior effectiveness and utility compared to other platforms. Despite being a highly valuable tool for sports and exercise-related investigations, its usage has been relatively minimal over the years. However, in various other domains, the platform has seen widespread application for tasks such as detecting hand gestures (Bora et al., 2023), pose estimation (Kim et al., 2023), action recognition (Ichsan Arsyi Putra et al., 2022), face recognition, and more.

3.1.5 Key points used for the features of the model using MediaPipe Framework

As a popular open-source platform for detecting skeletal points from human poses in images and videos, MediaPipe provides a flexible framework for building multi modal perception pipelines, which can process both real-time and offline data. MediaPipe framework can be used to identify 33 landmarks from a human skeleton. However, to calculate the mathematical relationship between these points and determining the knee

angle of a patient's legs, the researchers have decided to focus on only 6 landmarks. This decision was made because the landmarks in the hip, knee, and ankle joints play a crucial role in accurately measuring the knee joint angle of both legs. The selected six landmarks, essential for calculating the knee joint angle, are outlined in Figure 3.2.

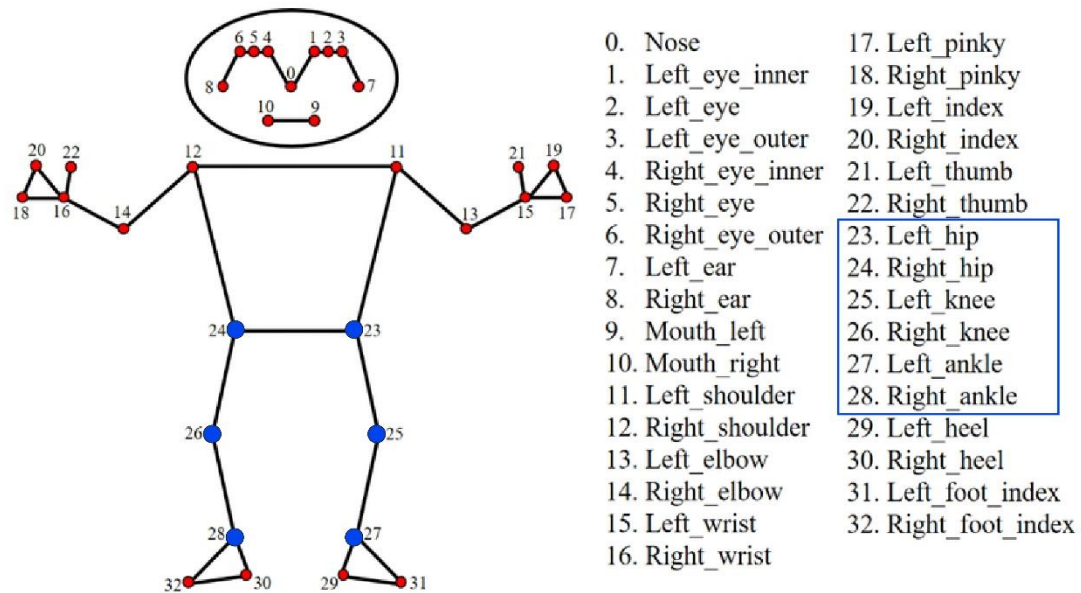


Figure 3.2 Six (6) Landmarks (blue colored) mainly focused on this research

The process of determining the knee joint angles of both legs of the patient by separately using these 6 landmarks can be explained as follows:

- **Left Knee Angle:** Left_hip, Left_knee, Left_ankle
- **Right Knee Angle:** Right_hip, Right_knee, Right_ankle

With these knee joint angles, it is possible to determine whether the patient achieves the target knee angle accurately during the prescribed exercises, as well as the duration taken to reach it.

3.1.6 Identifying clinically accepted parameters for Knee Range of Motion

Knee range of motion (ROM) serves as a critical clinical parameter in assessing knee function and overall musculoskeletal health. Achieving full ROM is imperative for facilitating daily activities and maintaining optimal biomechanical functionality. The knee ROM refers to the extent of movement that the knee joint can achieve in various directions. It is a crucial measure of the flexibility and functional capacity of the knee,

which is essential for performing daily activities and maintaining musculoskeletal health (Hancock et al., 2018).

In scientific terms, knee ROM is determined by the anatomical structure of the knee joint, including the articulation between the femur (thigh bone), tibia (shin bone), and patella (kneecap), as well as the surrounding ligaments, tendons, and muscles. The knee joint is classified as a hinge joint, allowing primarily flexion (bending) and extension (straightening) movements, with a limited degree of rotation.

Measurement of knee ROM involves assessing the angles formed between the femur and tibia during movement. Flexion refers to the bending of the knee joint, where the angle between the two bones decreases, while extension involves straightening the knee joint, increasing the angle. These movements occur in the sagittal plane, with flexion typically ranging from 0 to 135 degrees and extension ranging from 0 to 10 degrees beyond the anatomically neutral position.

There are three different ways to assess knee range of motion:

- **Active Knee ROM:** This refers to the range of motion achieved when the knee muscles contract without external assistance, allowing the knee to bend and straighten on its own.
- **Passive Knee ROM:** This measures how far the knee can bend and straighten when moved by an external force, typically another person, with the leg and knee muscles completely relaxed.
- **Active Assisted ROM:** This indicates the range of motion achievable when the knee is weak or painful, with some assistance provided, such as from a therapist. In this case, the knee muscles are working to move the leg, but with additional help.

It is common for the passive knee range of motion to be slightly greater than the active range of motion. In a healthy, pain-free joint, these three ranges of motion will be similar. However, in cases of knee injury, pain, or conditions like arthritis, the range of motion may be reduced, with active range of motion often being the most affected.

Normal knee range of motion refers to the degree of bending and straightening achievable by the knee joint. Additionally, a small amount of rotation can occur when the knee is bent. Figure 3.3 depicts the external and internal rotations of the leg when the knee is bent.

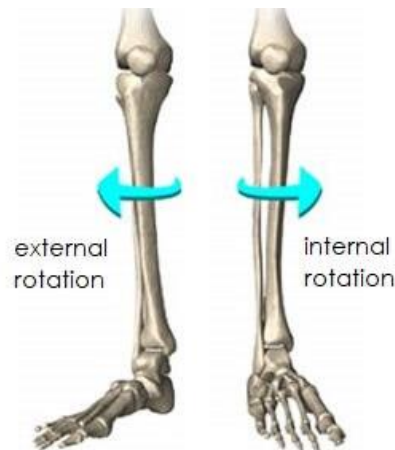


Figure 3.3 The external and internal rotations of the leg when the knee is bent

Normal active knee range of motion is as follows:

- **Knee Flexion ROM:** 135 degrees (fully bent)
- **Knee Extension ROM:** 0 degrees (fully straight)
- **Internal Knee Rotation ROM:** 10 degrees
- **External Knee Rotation ROM:** 30-40 degrees

Normal passive knee range of motion is:

- **Passive Knee Flexion ROM:** Up to 150 degrees, depending on the size of the leg. The limit is reached when the calf pushes onto the back of the thigh.
- **Passive Knee Extension ROM:** Up to 10 degrees of hyperextension is considered normal.

Normal functional range of motion (ROM) refers to the amount of movement required for typical daily activities such as walking, climbing stairs, and squatting down.

For the knee joint, most functional activities necessitate up to 120 degrees of knee flexion, rather than the full 135 degrees. However, virtually all functional activities require full knee extension.

The normal knee ROM required for activities of daily living includes:

- **Walking:** 0-65 degrees
- **Climbing upstairs:** 0-85 degrees
- **Descending stairs:** 0-90 degrees
- **Sitting down:** 0-90 degrees
- **Standing up from sitting:** 0-95 degrees
- **Tying your shoelace:** 0-105 degrees
- **Picking an object up from the floor:** 0-75 degrees
- **Riding a bike:** 0-115 degrees
- **Squatting:** At least 115 degrees
- **Sitting cross-legged:** At least 115 degrees

As demonstrated, even with a slightly limited knee flexion range of motion, most usual activities can still be performed. However, a small loss in knee extension range of motion can significantly impact functional ability.

Clinically, knee range of motion (ROM) is assessed using various methods, including goniometry, visual inspection, and advanced imaging techniques such as magnetic resonance imaging (MRI) or motion analysis systems. Goniometry involves utilizing a goniometer, a device specifically designed to measure joint angles, to quantify the degree of knee flexion and extension. Visual inspection enables clinicians to observe the range and quality of knee movement during functional activities.

Among these two methods, the most common and accurate method for measuring knee range of motion is by using a goniometer. This device resembles a specially designed protractor with a circle representing 360 degrees of motion at its centre. It consists of two extending arms: a stationary arm connected to the protractor and a moveable arm (Hancock et al., 2018).

When measuring the knee range of motion, it is essential to identify three specific landmarks on the leg:

- **Knee: Lateral Epicondyle** – the midpoint of the outer knee
- **Ankle: Lateral Malleolus** – the bony lump on the outer side of the ankle
- **Hip: Greater Trochanter** – a bony lump located just below the hip joint on the outer side of the thigh.

These three landmarks on the leg can be shown in Figure 3.4 below.

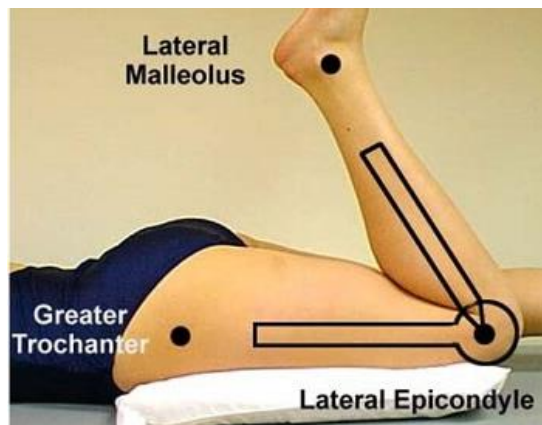


Figure 3.4 Specific landmarks on the leg to measure the knee range of motion

To locate the greater trochanter, first, identify the midpoint of the bony ridge on the top of the pelvis (the iliac crest), and then move your fingers downward approximately 15-20 cm until you feel a bony lump. The greater trochanter is situated more towards the back of the thigh due to the larger muscle bulk of the quadriceps on the front of the thigh. You may need to press through the soft tissues of the thigh to locate it accurately. Figure 3.5 below illustrates how clinicians and physiotherapists accurately measure knee range of motion using a goniometer.



Figure 3.5 Measuring knee range of motion using a goniometer

Ideally, a trained individual should perform the measurements, as obtaining an accurate measurement of knee ROM can be challenging when attempting to do it oneself. However, the goniometer has limitations, as it does not allow clinicians to analyze knee range of motion (ROM) and track the knee joint during activities such as walking or performing maximum squats. For this reason, motion capture devices are primarily utilized to analyze a patient's gait and assess the condition of the joints and bones.

Maintaining adequate knee ROM is crucial for optimal musculoskeletal function and performance in activities of daily living, such as walking, running, climbing stairs, and squatting. Limited knee ROM can lead to functional impairments, pain, instability, and an increased risk of musculoskeletal injuries. Therefore, assessing and addressing deficits in knee ROM through targeted rehabilitation interventions are essential for restoring function and preventing complications in individuals with knee pathology or injury.

3.1.7 Calculating the Euclidean distance between two points in a human skeleton using MediaPipe

In biomechanical analysis and motion tracking studies, accurately measuring the distances between specific points in a human skeleton is essential for assessing movement patterns and joint kinematics. One widely used method for quantifying these distances is through the calculation of Euclidean distances. The Euclidean distance, also known as straight-line distance, represents the shortest path between two points in a Euclidean space, such as the human body. In the context of human skeletal analysis, determining the Euclidean distance between key anatomical landmarks provides valuable insights into the spatial relationships and relative positions of various body segments. In this study, we focus on utilizing MediaPipe, to calculate the Euclidean distance between two points in the human skeleton.

The Euclidean distance is a measure of the straight-line distance between two points in Euclidean space. It is derived from the Pythagorean theorem, which relates the lengths of the sides of a right triangle. The Pythagorean theorem states that the square of the hypotenuse of a right-angled triangle is equal to the sum of the squares of its two other sides that will clearly show on the Figure 3.6 in detail.

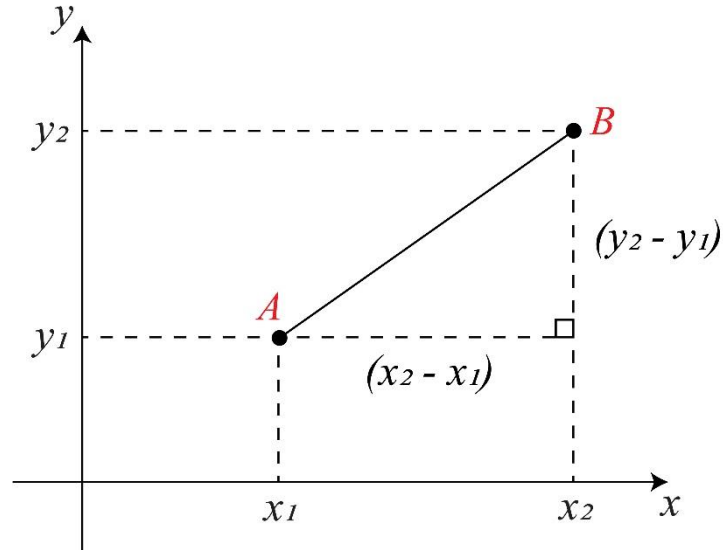


Figure 3.6 Euclidean distance on a 2D plane

As per the diagram above, we can get the distance between the two points A and B using the following formula.

$$Distance = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

where:

- *Distance* represents the Euclidean distance between points A and B.
- (x_1, y_1) and (x_2, y_2) are the coordinates of points A and B respectively.

This formula entails computing the difference between the coordinates of two points along each dimension, squaring these differences, summing them together, and finally, taking the square root of the sum to yield the distance. This distance metric is fundamental for precisely quantifying the spatial relationships between points, particularly in analyzing the distances between the hip, knee, and ankle joints using the MediaPipe framework.

3.1.8 Calculating the angle between three points in a human skeleton using MediaPipe

In biomechanical analysis and human motion tracking, accurately measuring joint angles is crucial for assessing movement patterns and kinematics. Specifically, in the context of knee joint analysis, determining the angle formed by three key anatomical

landmarks—namely, the hip, knee, and ankle joints—provides critical insights into joint function and movement dynamics. Utilizing advanced technology such as MediaPipe, a versatile framework for pose estimation, enables precise calculation of the angle between these three points in the human skeleton.

The angle between three points A, B, and C in a Euclidean space can be determined using the law of cosines formula. This fundamental formula in geometry allows for the calculation of various unknowns in a triangle, including missing sides or angles, as well as the classification of triangles as acute, obtuse, or right-angled. In this study, we will utilize the law of cosines to compute the angle formed by three two-dimensional skeletal points representing the hip, knee, and ankle joints. The law of cosine is shown in Figure 3.7.

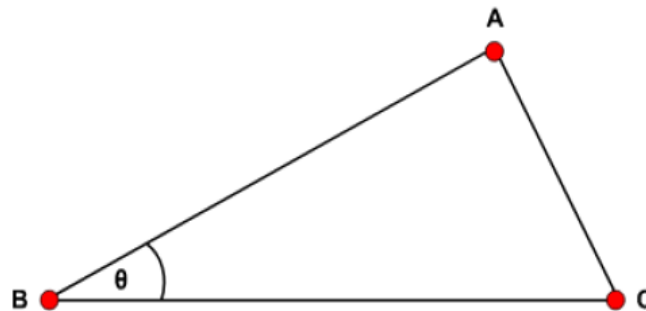


Figure 3.7 A triangle depicting the Law of Cosines

Let's denote the vectors AB and BC as follows:

$$AB = (B - A)$$

$$BC = (C - B)$$

The dot product (also known as scalar product) of these two vectors is given by:

$$AB \cdot BC = \|AB\| \|BC\| \cos(\theta)$$

where:

- $\|AB\|$ and $\|BC\|$ represent the magnitudes (or lengths) of vectors AB and BC respectively.
- θ is the angle between the two vectors.

The dot product of two vectors can be calculated as the sum of the products of their corresponding components. In this case, if $A = (x_1, y_1)$, $B = (x_2, y_2)$, and $C = (x_3, y_3)$, then:

$$AB \cdot BC = (x_2 - x_1)(x_3 - x_2) + (y_2 - y_1)(y_3 - y_2)$$

The magnitudes of the vectors AB and BC can be calculated using the Euclidean distance formula.

$$\| AB \| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\| BC \| = \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}$$

Substituting these into the dot product equation gives us the cosine of the angle θ .

$$\cos(\theta) = \frac{(AB \cdot BC)}{(\| AB \| \| BC \|)}$$

After plugging in the values from the Euclidean distance formula into the modified formula of the law of cosines, the angle θ can be obtained by applying the arccosine function.

$$\theta = \cos^{-1} \left[\frac{(AB \cdot BC)}{(\| AB \| \| BC \|)} \right]$$

It is important to note that this angle will be represented in radians. To convert the angle value from radians to degrees, the following formula will be applied as follows.

$$\text{Angle in degrees} = \left[\theta \times \frac{180}{\pi} \right]$$

where:

- θ is in radians.

By applying the above formula, we can accurately compute the angle formed by the hip, knee, and ankle joints in the human skeleton using MediaPipe. This calculated angle provides valuable insights into the biomechanics of the knee joint during various movements and exercises. Understanding the precise angle between these key

anatomical landmarks aids in evaluating joint function, identifying abnormalities, and informing targeted interventions for conditions such as knee osteoarthritis.

Moreover, this methodology enables clinicians and researchers to objectively assess knee joint kinematics, contributing to advancements in rehabilitation techniques and personalized treatment approaches. Thus, the application of this mathematical approach using MediaPipe facilitates a comprehensive understanding of knee joint angles and their implications for human movement and health.

3.1.9 Optimal camera angle for accurate detection of Knee Joint Angle in patients

In biomechanics and rehabilitation, it is vital to accurately evaluate the angle of the knee joint to monitor the progress of patients undergoing therapy. This is especially important for those who suffer from knee osteoarthritis. One key factor influencing the accuracy of knee joint angle detection is the camera angle used during data acquisition. Therefore, the researchers investigate the optimal camera angle that maximizes precision in detecting the knee joint angle of patients performing rehabilitation exercises.

In this study, there are several factors need to be considered when determining the appropriate camera angle for detecting the knee joint angle:

- **Visibility of Anatomical Landmarks:** The camera angle should ensure clear visibility of the hip, knee, and ankle joints, as well as relevant anatomical landmarks used for angle calculation.
- **Minimization of Foreshortening:** Foreshortening occurs when objects appear shorter or compressed in the image due to the camera angle. Minimizing foreshortening is essential for accurate measurement of joint angles.
- **Consistency and Stability:** The camera angle should be consistent and stable throughout the recording session to ensure reliable data collection and analysis.
- **Comfort and Safety of the Patient:** The camera angle should prioritize the comfort and safety of the patient during exercise performance, minimizing any discomfort or risk of injury.

Based on the preliminary investigations and expert recommendations, the optimal camera angle for detecting the knee joint angle in patients is determined to be a lateral view positioned perpendicular to the sagittal plane of the body. Figure 3.8 show the proper way to arrange the patient's body and the camera, and its maximum distance (3m) are clearly indicated as a top view. This camera angle provides a clear and unobstructed view of the hip, knee, and ankle joints, minimizing foreshortening effects and ensuring accurate measurement of joint angles. Additionally, maintaining a consistent and stable camera position throughout the recording session enhances the reliability of the data obtained.

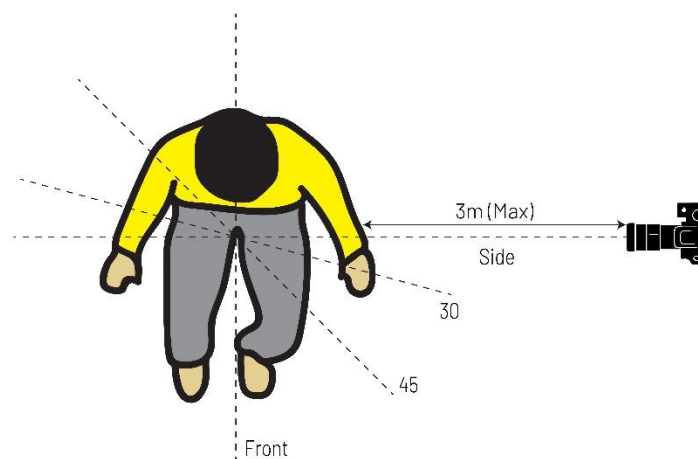


Figure 3.8 Optimal camera angle for accurate detection of Knee Joint Angle

The optimal camera angle, determined to be a lateral view perpendicular to the sagittal plane of the body, ensures clear visibility of anatomical landmarks and minimizes foreshortening effects, thereby maximizing the accuracy of joint angle measurement.

3.1.10 Biomechanical model development

The development of a biomechanical model using skeleton landmarks extracted through MediaPipe offers a promising approach to measure knee angles during rehabilitation exercises. This study aims to elucidate the process of biomechanical model development and its application in evaluating the correctness of exercises performed by patients, while also assessing their achievement of daily knee angle goals.

The development of the biomechanical model involves several key steps:

1. **Skeleton Landmark Detection:** Utilizing the MediaPipe framework, skeletal landmarks corresponding to key anatomical points, including the hip, knee, and ankle joints, are detected from video inputs of patients performing rehabilitation exercises. These landmarks serve as reference points for subsequent angle calculations.
2. **Angle Calculation:** With the identified skeletal landmarks, the knee joint angle is calculated using established mathematical principles, such as trigonometry. By determining the angular relationships between the hip, knee, and ankle joints, the precise knee angle during exercise execution is quantified.
3. **Correctness Assessment:** The calculated knee angle is compared against predefined thresholds or reference values representing the correct execution of rehabilitation exercises. Any deviations from the expected angle range indicate potential errors in exercise performance, allowing for real-time feedback to patients and clinicians.
4. **Daily Goal Evaluation:** In addition to assessing exercise correctness, the biomechanical model facilitates the evaluation of patients' achievement of daily knee angle goals. By setting specific angle targets based on clinical guidelines or individual rehabilitation plans, the model tracks patients' progress towards meeting these goals over time.

By leveraging skeleton landmarks and mathematical calculations, this model offers a comprehensive approach to assessing exercise correctness and monitoring patients' achievement of daily knee angle goals. As an integral component of rehabilitation practice, the biomechanical model contributes to enhancing the precision and efficacy of exercise interventions for patients with knee osteoarthritis, ultimately improving their functional outcomes and quality of life.

3.1.11 Performance parameter model development

In order to monitor patients' progress and customize interventions accordingly, it is essential to develop a performance parameter model that systematically measures and records critical exercise parameters. This explanation explains the process of developing such a model and how it can capture essential data related to exercise performance in patients with knee osteoarthritis.

The development of the performance parameter model involves the following steps:

1. **Parameter Identification:** A comprehensive set of parameters relevant to exercise performance and rehabilitation progress in knee osteoarthritis patients is identified by developing the performance parameter model. These parameters encompass various aspects of exercise execution, including temporal, positional, and biomechanical metrics.
2. **Data Collection Methodology:** The model specifies the data collection methodology, outlining the procedures for capturing each parameter during exercise sessions. This is obtained by using the MediaPipe framework.
3. **Parameter Measurement:** During exercise sessions, the identified parameters are systematically measured and recorded in real-time. This includes documenting the exercising date and time, patient's body position, knee angle corresponding to the relevant exercise, duration of knee angle holding, relaxation intervals, repetition count, total exercise duration, and the distance between two legs.
4. **Data Storage and Management:** The collected data of parameters are stored in a structured format, typically in a CSV (Comma-Separated Values) file, for efficient storage and retrieval. Each patient's data is saved under their respective name or identifier, allowing for personalized tracking of rehabilitation progress over time.
5. **Quality Assurance:** The performance parameter model incorporates mechanisms for ensuring data accuracy and reliability. This may involve validation checks, data cleaning procedures, and error handling protocols to minimize inaccuracies or inconsistencies in the collected data.

By systematically capturing exercise-related parameters, the model enables comprehensive monitoring of patients' performance and progress throughout their rehabilitation journey. Clinicians can leverage the collected data to assess the effectiveness of specific exercises, identify areas for improvement, and adjust treatment plans accordingly. Moreover, researchers can utilize the aggregated data to conduct

longitudinal analyses, evaluate treatment outcomes, and inform future research directions in knee osteoarthritis rehabilitation.

3.1.12 Model Evaluation

The evaluation of biomechanical and performance parameter models in knee osteoarthritis (OA) rehabilitation involves soliciting expert opinions from domain specialists, particularly physiotherapists, to assess the utility, effectiveness, and feasibility of these models in clinical practice. This study elucidates the methodology and rationale behind utilizing expert opinion analysis to evaluate the biomechanical and performance parameter models in knee OA rehabilitation. Domain experts in the medical field, particularly physiotherapists specializing in knee OA rehabilitation, are selected as evaluators for the models. Experts are chosen based on their clinical experience, expertise in knee OA management, and familiarity with rehabilitation protocols.

Criteria for evaluating the biomechanical and performance parameter models are established in collaboration with the expert panel. These criteria encompass various dimensions, including:

- **Accuracy:** The ability of the models to accurately measure knee joint angles and capture exercise-related parameters.
- **Clinical Relevance:** The extent to which the models align with established clinical guidelines and rehabilitation protocols for knee OA.
- **Usability:** The ease of use and practicality of implementing the models in real-world clinical settings.
- **Patient Engagement:** The models' ability to engage and motivate patients in adhering to rehabilitation exercises.
- **Data Interpretation:** The clarity and comprehensibility of the data generated by the models for clinical decision-making.

Experts are presented with the biomechanical and performance parameter models, along with relevant documentation and demonstration materials. They are then asked to provide structured feedback and opinions on the models based on the established

evaluation criteria. The feedback and opinions provided by the expert panel are systematically analyzed to identify common themes, patterns, and areas of consensus or divergence regarding the models' efficacy and usability. The findings from the expert opinion analysis are synthesized to generate comprehensive insights into the strengths, limitations, and potential enhancements of the biomechanical and performance parameter models related to this study.

The utilization of expert opinion analysis in evaluating the biomechanical and performance parameter models offers several advantages:

- **Leveraging Clinical Expertise:** Physiotherapists bring valuable clinical insights and practical experience to the evaluation process, ensuring that the models are assessed from a clinically relevant perspective.
- **Validation of Clinical Utility:** Expert opinions validate the clinical utility and applicability of the models in real-world knee OA rehabilitation practice, enhancing their credibility and acceptability among healthcare professionals.
- **Identification of Improvement Areas:** Feedback from experts facilitates the identification of areas for improvement and refinement in the models, guiding future iterations and enhancements to better meet the needs of clinicians and patients.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Introduction

In this phase of the study, patients are evenly distributed into two groups: one group utilizes the Virtual Physiotherapy Guide developed by the researchers, while the other group undergoes traditional rehabilitation methods without the aid of a guide. The patients selected for this study fall within a specific age range, and detailed instructions are provided to them regarding the correct execution of the prescribed exercises. Prior to commencing the exercises, the Knee Injury and Osteoarthritis Outcome Score (KOOS) is assessed for each patient to establish a baseline. Subsequently, after completing the designated exercises, the corresponding KOOS score is re-evaluated and compared against the baseline measurement. This comparative analysis serves as a quantitative indicator of the effectiveness of the Virtual Physiotherapy Guide in improving patient outcomes and facilitating optimal rehabilitation.

4.2 Patient Inclusion Criteria for Virtual Physiotherapy Guider Study in Knee Osteoarthritis

This section outlines the patient inclusion criteria implemented in the study evaluating the efficacy of the Virtual Physiotherapy Guider for Knee Osteoarthritis using Computer Vision technology.

4.2.1 Patient Inclusion Criteria

Patients eligible for inclusion in this study met the following criteria:

1. **Age Range:** Patients aged between 35 and 45 years.
2. **Diagnosis:** Patients diagnosed with unilateral primary knee osteoarthritis.
3. **Technology Access:** Patients must possess and have the capability to use a laptop equipped with an integrated webcam.

4.2.2 Patient Exclusion Criteria

Patients were excluded from the study if they met any of the following criteria:

1. **History of Knee Replacement Surgery:** Patients who had undergone knee replacement surgery.
2. **Medical Comorbidities:** Patients with significant medical comorbidities that may interfere with rehabilitation exercises, including:
 - Previous Stroke
 - Cardiac Diseases
 - Severe Hypertension
 - Severe Obesity
 - Chronic Obstructive Pulmonary Disease
 - Low Back Pain
 - Chronic Pain
 - Depression
 - Visual or Hearing Impairments
3. **Previous Exercise Instruction:** Patients who had been instructed to perform specific exercises relevant to the study prior to the recruitment period.

The patient inclusion and exclusion criteria were carefully designed to ensure the study's integrity, safety, and effectiveness. By including patients within a specific age range and with a confirmed diagnosis of unilateral primary knee osteoarthritis, the study focuses on a homogeneous patient population. Additionally, the requirement for laptop access with an integrated webcam ensures consistent participation and data collection using the Virtual Physiotherapy Guider. Conversely, the exclusion of patients with significant medical comorbidities or prior exposure to relevant exercises minimizes confounding factors and ensures the study's validity.

4.3 Randomization

Randomization was employed to assign eligible patients into two distinct groups: the virtual physiotherapy guider-directed exercise guidance group (referred to as the A-

group) and the conventional education handout self-directed exercise guidance group (referred to as the B-group), with each group comprising three participants ($n = 3$ each). This random allocation ensures that each participant has an equal chance of being assigned to either group, thereby minimizing selection bias. The design of these two groups is visually depicted in Figure 4.1, illustrating the allocation process and highlighting the distinct pathways for each group throughout the study protocol.

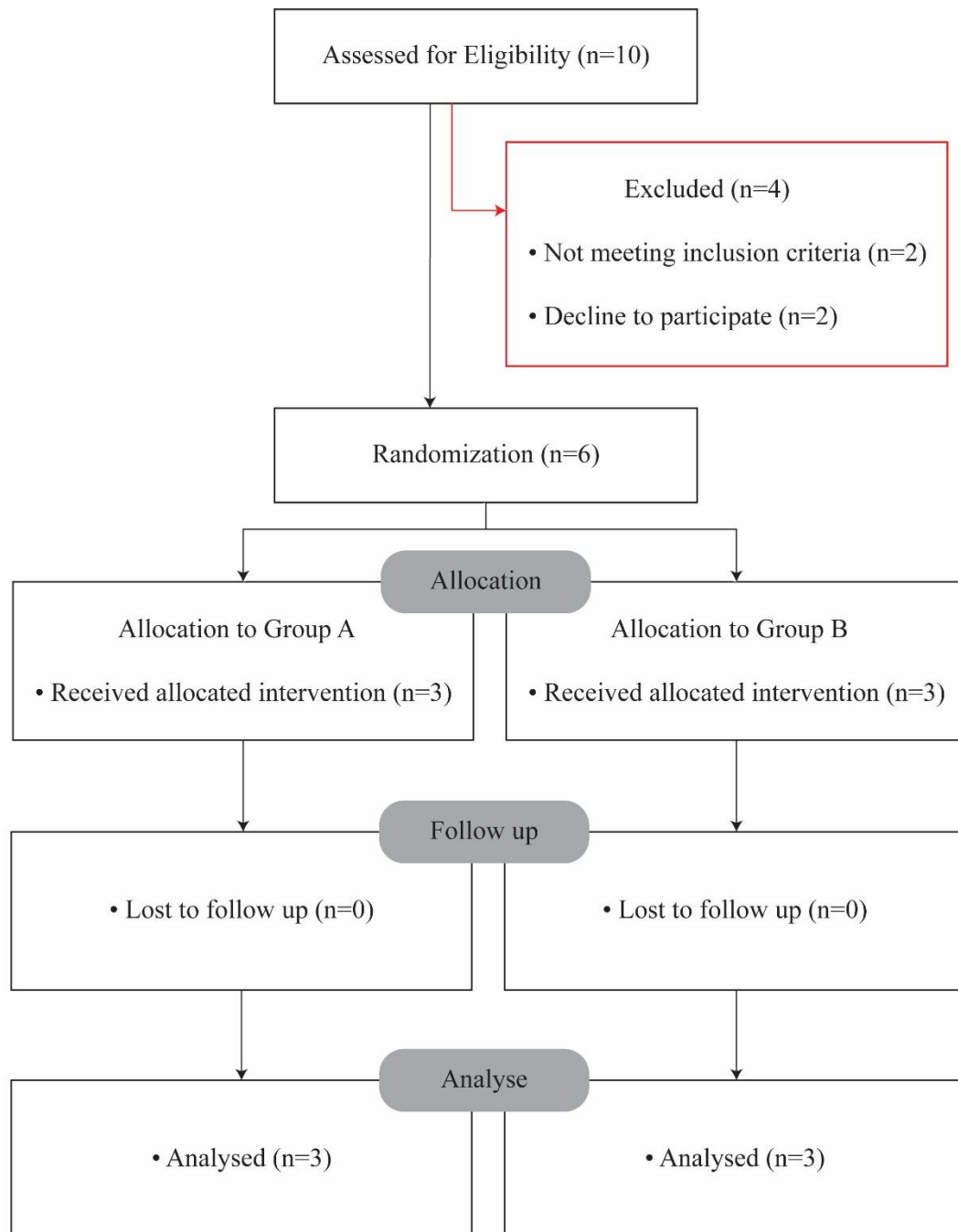


Figure 4.1 The pathway of each group throughout the study protocol

Demographic information, encompassing age, gender, affected side, body mass index (BMI), onset of knee pain, educational attainment, range of motion (ROM), and the Knee Injury and Osteoarthritis Outcome Score (KOOS), was meticulously documented for all participants. Both groups received identical comprehensive information concerning knee osteoarthritis (OA) pathology and exercise instructions to ensure uniformity in knowledge dissemination. The gathered data, including the aforementioned demographics, are presented in Table 4.2, offering a concise overview of the participant characteristics and clinical parameters under investigation.

4.4 Intervention

Following the assignment to their respective groups, patients underwent exercise instruction, with Group A receiving guidance through the virtual physiotherapy guider, while Group B received traditional handouts. The standardized exercise regimen comprised three distinct exercises: Straight Leg Raise, Knee Extension, and Reverse Lunge. These exercises were selected based on their effectiveness in targeting key muscle groups and promoting knee joint mobility and stability.

4.5 Guidelines for the relevant exercises

4.5.1 Straight Leg Raise

To perform the Straight leg raise exercise, begin by lying on your back with your hips aligned and legs extended comfortably on the floor. Flex the knee of your unaffected leg to a 90-degree angle, keeping the foot firmly planted on the ground. Engage your quadriceps muscles to stabilize the straight leg. Inhale deeply as you slowly raise the straight leg approximately six inches off the ground, ensuring controlled movement. Hold this position for three seconds, maintaining stability and control. Exhale slowly as you lower the leg back to the floor, continuing to engage the muscles. Relax briefly before repeating the movement for a total of 10 repetitions. This exercise effectively targets the quadriceps and promotes knee joint strength and stability.

4.5.2 Knee Extension

To perform the Knee Extension exercise, start by sitting in a chair with your back and hips against the chair's backrest. Ensure that your knees are bent at a 90° angle, with only your toes lightly touching the ground. If the chair height is inadequate, adjust it by adding cushions or placing a rolled towel under your knees for proper alignment.

Position your arms either resting on your thighs or by your sides, and maintain a shoulder-width distance between your feet.

Execute the movement in a controlled manner, following a 1-2-Up count. Extend your left leg outward, aiming to achieve maximum extension while maintaining proper form and without experiencing any pain. Pause for one second when your leg reaches its highest point or when it is parallel to the ground. Then, in a 1-2-3-Down count, slowly lower your leg back to the starting position at a 90° angle.

Complete 20 repetitions of this exercise for each leg, constituting one set. Rest for 30 seconds before proceeding to perform the second set. This exercise targets the quadriceps muscles and helps improve knee joint strength and stability when performed correctly and consistently.

4.5.3 Reverse Lunge

To execute the Reverse Lunge exercise correctly, begin by standing upright with your feet hip-width apart. Engage your core muscles to maintain stability throughout the movement. Take a controlled step backward with one foot, ensuring to lower your body until both knees are bent at approximately 90-degree angles. Keep your front thigh parallel to the floor, with your knee positioned directly above your ankle.

As you lower your body, focus on maintaining proper alignment by keeping your chest lifted and your back straight. Simultaneously, allow your back knee to descend toward the floor while lifting the heel off the ground. This position emphasizes engagement in the muscles of the front leg, particularly the quadriceps and glutes.

Hold this lowered position briefly to enhance the challenge to your lower body muscles. Then, smoothly return to the starting position by exerting force through the heel of your front foot. As you rise, concentrate on pushing upward and squeezing the glute muscles of your front leg while simultaneously unbending the back leg. Ensure controlled movement throughout the exercise to maximize effectiveness and minimize the risk of injury.

Repeat the Reverse Lunge movement for the desired number of repetitions on each leg, alternating between legs to achieve balanced muscle engagement. This exercise targets various muscle groups, including the quadriceps, hamstrings, glutes, and calves,

contributing to improved lower body strength, stability, and functional movement patterns.

4.6 Outcome measurements

Outcome measurements were conducted after 4 weeks of self-guided practice by an independent clinician who was blinded to the intervention assignment. The primary outcome assessed the patients' ability to execute the prescribed exercises accurately. Patients were instructed to perform each of the three exercises ten times, with correct performance defined as accurately completing at least 8 out of 10 repetitions.

Secondary measurements encompassed clinical outcomes related to range of motion (ROM) and Knee injury and Osteoarthritis Outcome Score (KOOS) categories, including symptoms, pain, activities of daily living, sports and recreational activities, and quality of life. ROM assessment was conducted with the patient in the supine position using a goniometer, a standard tool for measuring joint angles and range of motion. This comprehensive evaluation aimed to gauge the effectiveness of the intervention in improving patients' exercise performance and clinical outcomes.

4.6.1 Calculate the KOOS score

The Knee Injury and Osteoarthritis Outcome Score (KOOS) is a validated tool utilized to assess patient-relevant outcomes following a knee injury, encompassing both short-term and long-term evaluations. It comprises five distinct domains: pain, symptoms, activities of daily living, sports and recreation function, and knee-related quality of life. The KOOS is a self-administered questionnaire designed to adhere to essential standards for outcome measures, enabling the comprehensive assessment of knee injury progression and treatment efficacy. This patient-reported outcome measure is characterized by its user-friendly format, allowing patients to complete it independently within approximately 10 minutes.

4.6.2 Scoring and Interpretation

The KOOS questionnaire assesses five patient-relevant dimensions: pain, symptoms, daily activities, sports and recreation, and quality of life. Respondents rate each dimension using a Likert scale ranging from 0 to 4, where 0 signifies no issues and 4 indicates extreme problems. The scores for each dimension are derived by summing the responses to relevant questions and then transformed to a standardized scale of 0 to 100.

Here, 0 denotes severe knee problems, while 100 signifies no knee-related issues. These scores represent the percentage of the maximum achievable score. Previously, each dimension allowed for up to two missing responses, with at least 50% of the items requiring a reply. The following formulas are used to calculate each subscale score:

- KOOS Symptoms:

$$100 - \left[\frac{\text{mean score symptoms} \times 100}{4} \right]$$

- KOOS Pain:

$$100 - \left[\frac{\text{mean score pain} \times 100}{4} \right]$$

- KOOS ADL:

$$100 - \left[\frac{\text{mean score ADL} \times 100}{4} \right]$$

- KOOS Sport/ Recreation:

$$100 - \left[\frac{\text{mean score sport/ recreation} \times 100}{4} \right]$$

- KOOS Quality of life:

$$100 - \left[\frac{\text{mean score QOL} \times 100}{4} \right]$$

4.6.3 KOOS survey for the Knee Osteoarthritis

This survey is designed to gather insights regarding the knee condition of patients, facilitating the evaluation of their knee-related experiences and functional abilities. Patients are required to respond to each question by selecting the most appropriate option from the provided scale of five choices: Never, Rarely, Sometimes, Often, and

Always. It is imperative that patients select only one option corresponding to their experience. In cases of uncertainty, patients are encouraged to provide their best response. The feedback obtained from this survey will play a crucial role in accurately monitoring the patients' knee health and activity performance.

- **Symptoms:** The responses to these questions should be based on the knee symptoms experienced by the respondent during the preceding week.
 - i. Do you have swelling in your knee?
 - ii. Do you feel grinding, hear clicking or any other type of noise when your knee moves?
 - iii. Does your knee catch or hang up when moving?
 - iv. Can you straighten your knee fully?
 - v. Can you bend your knee fully?
- **Stiffness:** The subsequent questions pertain to the extent of joint stiffness encountered by individuals in their knee over the past week. Joint stiffness is characterized by a perception of constraint or sluggishness in the ease of movement within the knee joint. It encompasses sensations of reduced flexibility or resistance during motion, often accompanied by discomfort or difficulty in articulation.
 - i. How severe is your knee joint stiffness after first wakening in the morning?
 - ii. How severe is your knee stiffness after sitting, lying or resting later in the day?
- **Pain:** These questions seek to assess what amount of knee pain have respondent experienced the last week during the following activities.
 - i. How often do you experience knee pain?
 - ii. Twisting/ pivoting on your knee?
 - iii. Straightening knee fully?
 - iv. Bending knee fully?
 - v. Walking on flat surface?
 - vi. Going up or down stairs?
 - vii. At night while in bed?

- viii. Sitting or lying?
- ix. Standing upright?

- **Function, Daily living:** The subsequent inquiries pertain to the respondent physical function, encompassing their capacity to navigate and attend to personal needs. Each item inquiries about the extent of impediments encountered in the past week attributable to knee-related factors. Participants are requested to quantify the level of challenge experienced during specified activities, shedding light on the impact of knee conditions on physical functionality and self-care capabilities over the specified period.

- i. Descending stairs?
- ii. Ascending stairs?
- iii. Rising from sitting?
- iv. Standing?
- v. Bending to floor/pick up an object?
- vi. Walking on flat surface?
- vii. Getting in/ out of car?
- viii. Going shopping?
- ix. Putting on socks/ stockings?
- x. Rising from bed?
- xi. Taking off socks/stockings?
- xii. Lying in bed (turning over, maintaining knee position)?
- xiii. Getting in/ out of bath?
- xiv. Sitting?
- xv. Getting on/ off toilet?
- xvi. Heavy domestic duties (moving heavy boxes, scrubbing floors, etc)
- xvii. Light domestic duties (cooking, dusting, etc)

- **Function, Sports and Recreational Activities:** The following questions concern the respondent physical function when being active on a higher level.

- i. Squatting?
- ii. Running?
- iii. Jumping?

- iv. Twisting/pivoting on your injured knee?
 - v. Kneeling?
- **Quality of life:** The quality-of-life questions in this survey aim to assess the overall impact of knee-related symptoms and limitations on various aspects of daily living. Participants are asked to reflect on their experiences over the past week and provide insights into how their knee condition has influenced their overall well-being, including emotional well-being, social interactions, and overall satisfaction with life. The following questions are related to the Quality-of-Life section.
 - i. How often are you aware of your knee problem?
 - ii. Have you modified your life style to avoid potentially damaging activities to your knee?
 - iii. How much are you troubled with lack of confidence in your knee?
 - iv. In general, how much difficulty do you have with your knee?

The following table presents the KOOS data collected prior to the commencement of exercises for the six participating patients.

KOOS score of patients before intervention taken on 20/12/2023	
Group A	Group B
Patient 1 (Age: 36) KOOS Symptoms: 64 KOOS Pain: 56 KOOS ADL: 59 KOOS Sports/Recreation: 50 KOOS QOL: 63	Patient 4 (Age: 42) KOOS Symptoms: 25 KOOS Pain: 31 KOOS ADL: 35 KOOS Sports/Recreation: 30 KOOS QOL: 44
Patient 2 (Age: 40) KOOS Symptoms: 43 KOOS Pain: 47 KOOS ADL: 54 KOOS Sports/Recreation: 40 KOOS QOL: 50	Patient 5 (Age: 38) KOOS Symptoms: 82 KOOS Pain: 83 KOOS ADL: 75 KOOS Sports/Recreation: 75 KOOS QOL: 69
Patient 3 (Age: 45) KOOS Symptoms: 36 KOOS Pain: 33 KOOS ADL: 40 KOOS Sports/Recreation: 30 KOOS QOL: 44	Patient 6 (Age: 41) KOOS Symptoms: 50 KOOS Pain: 53 KOOS ADL: 54 KOOS Sports/Recreation: 50 KOOS QOL: 69

Table 4.1 KOOS score of patients before intervention taken on 20/12/2023

4.7 Patients' Demographics table

The demographic table provides an overview of key characteristics pertaining to the patients involved in the study, including age, gender, side of knee affected, body mass index (BMI), onset of knee pain, education level, range of motion (ROM), KOOS score. The table below presents demographic data for the six patients participating in the study.

	A group (n=3)	B group (n=3)
Age ^a (y)	40.3	40.3
Gender ^b (%) • Male • Female	1 (33.4) 2 (66.7)	1 (33.4) 2 (66.7)
Side ^b (%) • Right • Left	2 (66.7) 1 (33.4)	1 (33.4) 2 (66.7)
BMI (kg/m ²)	32.0	31.5
Time in years since the onset of knee pain ^b (%) • < 2 years • ≥ 2 years	1 (33.4) 2 (66.7)	1 (33.4) 2 (66.7)
Education level ^b (%) • < Secondary education • ≥ Secondary education • ≥ Bachelor	0 2 (66.7) 1 (33.4)	0 2 (66.7) 1 (33.4)
Occupation ^b (%) • Housewife • Professional and administration • Employee	1 (33.3) 1 (33.3) 1 (33.3)	0 2 (66.7) 1 (33.4)
ROM ^a (°)	120 ⁰	120.3 ⁰
KOOS ^a • Symptoms • Pain • Activities of daily living • Sports and recreational activities • Quality of life	47.7 45.3 51.0 42.0 52.3	52.3 55.7 54.7 51.7 60.7
<i>BMI: Body Mass Index, ROM: Range of Motion, KOOS: Knee Injury and Osteoarthritis Outcome Score,</i> ^a Presented as mean ^b Presented as number of patients		

Table 4.2 Demographics table of the participating patients

4.8 Results

The following table presents the KOOS data collected after the participating patients completed their exercises.

KOOS score of patients after intervention taken on 17/01/2024	
Group A	Group B
Patient 1 (Age: 36) KOOS Symptoms: 71 KOOS Pain: 67 KOOS ADL: 71 KOOS Sports/Recreation: 65 KOOS QOL: 69	Patient 4 (Age: 42) KOOS Symptoms: 36 KOOS Pain: 39 KOOS ADL: 44 KOOS Sports/Recreation: 35 KOOS QOL: 44
Patient 2 (Age: 40) KOOS Symptoms: 64 KOOS Pain: 67 KOOS ADL: 69 KOOS Sports/Recreation: 60 KOOS QOL: 63	Patient 5 (Age: 38) KOOS Symptoms: 86 KOOS Pain: 86 KOOS ADL: 82 KOOS Sports/Recreation: 80 KOOS QOL: 69
Patient 3 (Age: 45) KOOS Symptoms: 50 KOOS Pain: 50 KOOS ADL: 53 KOOS Sports/Recreation: 40 KOOS QOL: 50	Patient 6 (Age: 41) KOOS Symptoms: 64 KOOS Pain: 64 KOOS ADL: 62 KOOS Sports/Recreation: 55 KOOS QOL: 69

Table 4.3 KOOS score of patients after intervention taken on 17/01/2024

• Table 1

Outcomes of Group – A, before intervention and after 4 weeks of the Virtual physiotherapy guider - directed exercise.

	Before intervention	After 4 weeks
ROM ^a (°)	120 ⁰	122.8 ⁰
KOOS ^a		
Symptoms	47.7	61.7
Pain	45.3	61.3
Activities of daily living	51.0	64.4
Sports and recreational activities	42.0	55
Quality of life	52.3	60.7
ROM: Range Of Motion, KOOS: Knee Injury and Osteoarthritis Outcome Score, ^a Presented as mean		

Table 4.4 (Table 1) Outcomes of Group – A, before intervention and after 4 weeks of the Virtual physiotherapy guider - directed exercise

• **Table 2**

Outcomes of Group – B, before intervention and after 4 weeks of the handout self - directed exercise.

	Before intervention	After 4 weeks
ROM ^a (°)	120.3 ⁰	121.9 ⁰
KOOS ^a		
Symptoms	52.3	62
Pain	55.7	63
Activities of daily living	54.7	62.7
Sports and recreational activities	51.7	56.7
Quality of life	60.7	60.7
ROM: Range Of Motion, KOOS: Knee Injury and Osteoarthritis Outcome Score, ^a Presented as mean		

Table 4.5 (Table 2) Outcomes of Group – B, before intervention and after 4 weeks of the handout self - directed exercise

• **Table 3**

Comparison of outcomes between the two groups after 4 weeks.

	Group – A (n=3)	Group – B (n=3)
ROM ^a (°)	122.8 ⁰	121.9 ⁰
KOOS ^a		
Symptoms	61.7	62.0
Pain	61.3	63.0
Activities of daily living	64.4	62.7
Sports and recreational activities	55.0	56.7
Quality of life	60.7	60.7
ROM: Range Of Motion, KOOS: Knee Injury and Osteoarthritis Outcome Score, ^a Presented as mean		

Table 4.6 (Table 3) Comparison of outcomes between the two groups after 4 weeks

• **Table 4**

Comparison of outcome exercise between the two groups.

	Group – A (n=3)	Group – B (n=3)
Patient performing total exercises ^a (%)		
• Correct	2 (66.7%)	1 (33.3%)
• Incorrect	1 (33.3%)	2 (66.7%)

Patient performing correct exercise 1–3 ^a (%)		
• Exercise 1	3 (100%)	2 (66.7%)
• Exercise 2	3 (100%)	2 (66.7%)
• Exercise 3	2 (66.7%)	1 (33.3%)
^a Presented as number of patients		

Table 4.7 (Table 4) Comparison of outcome exercise between the two groups

Six patients meeting the inclusion criteria participated in this study, with 3 allocated to the A-group and 3 to the B-group. Table 4.2 presents the demographic details of patients in this randomized controlled trial (RCT). Patients were matched for age, gender, side, body mass index (BMI), and onset of knee pain. The average age of patients in both groups was 40.3 years, with females constituting 66.7% of each group. The average BMI was 32.0 in the A group and 31.5 in the B group. Education levels were also similar between groups, with the majority having received at least secondary education. Additionally, patients were matched for range of motion (ROM) and Knee injury and Osteoarthritis Outcome Score (KOOS) categories.

While there was no significant improvement in knee ROM after the 4-week exercise regimen in either group, both groups showed improved functional outcomes based on KOOS assessments (Tables 4.4, 4.5). Comparing KOOS scores between groups after the regimen revealed that patients in the A group had superior overall functional outcomes, including better daily life activities, sports and recreation abilities, higher satisfaction, and expectation levels (Table 4.6). Notably, patients in the A group reported significantly better quality of life compared to those in the B group.

Furthermore, the accuracy of performing the prescribed exercises at the end of the regimen was evaluated. Patients in the A group demonstrated higher accuracy in completing all exercises compared to the B group (Table 4.7). Specifically, the A group showed significantly better performance in exercises 1 and 2, with no significant difference in exercise 3.

4.9 Discussion

This study delved into the efficacy of an interactive and user-friendly virtual physiotherapy guider in delivering exercise instruction and providing real-time feedback to patients, ultimately aiming to enhance clinical outcomes and satisfaction. Conducted as a Randomized Control Trial (RCT), researcher findings revealed that patients undergoing rehabilitation guided by the Virtual Physiotherapy Guider for Knee Osteoarthritis using Computer Vision & Machine Learning exhibited enhanced rehabilitation accuracy, improved activities of daily living, quality of life, and increased ability to engage in sports and recreational activities compared to those relying on conventional educational handouts.

Notably, the virtual guider also bolstered patient satisfaction and expectation regarding knee performance when juxtaposed with traditional handouts. Muscle-strengthening exercises, such as those prescribed in this study, have been established to ameliorate pain and enhance functional capacity in knee OA rehabilitation, a recommendation echoed by the Osteoarthritis Research and Society International (OARSI). The prescribed exercise regime must be performed accurately and regularly for optimal rehabilitation, with the guider identifying key exercises including straight leg raise, knee extension, and reverse lunge.

The higher accuracy rate observed in patients completing exercises with the Virtual Physiotherapy Guider for Knee Osteoarthritis suggests the potential for sustained rehabilitation adherence in the long term. This study findings align with previous studies indicating that patients with musculoskeletal conditions tend to exhibit better adherence to home exercises when utilizing virtual physiotherapy guidance compared to conventional paper handouts. It is worth noting that while this study trial focused on non-surgical knee OA patients, the application of a virtual guider during postoperative rehabilitation has shown promise in reducing opiate usage for pain management and improving functional outcomes.

The developed Virtual Physiotherapy Guider for Knee Osteoarthritis using Computer Vision & Machine Learning emerges as a valuable tool for OA patients, facilitating exercise motivation, cost-effectiveness, and enhancing rehabilitation accuracy. However, certain limitations must be acknowledged. Firstly, the ability of patients to efficiently use laptops may not be representative of a broader elderly population.

Secondly, the inability to blind participants to their intervention could potentially bias self-reported outcomes. Additionally, the gender bias observed in this study, with a majority of female participants, is consistent with trends in OA patients but warrants consideration in future research through randomized gender stratification. Moreover, the absence of reference values to discern significant improvement among patients underscores the need for further investigation. Lastly, a long-term follow-up study is imperative to validate the sustained positive effects of mobile application usage on exercise adherence and patient health status.

The emergence of electronic health interventions, particularly virtual physiotherapy guiders, presents a promising avenue for supporting patient adherence to home exercise programs and improving physical function in OA patients. The interactive features embedded within these applications, such as exercise tracking and progress monitoring, hold potential to enhance rehabilitation accuracy and ultimately contribute to an improved quality of life for patients.

CHAPTER 5

CONCLUSION AND RECOMMENDATION

5.1 Introduction

This chapter serves as the culmination of the research findings, providing a comprehensive discussion of the conclusions drawn from the conducted study. Section 5.2 will delve into the key conclusions derived from the research outcomes, shedding light on the implications and significance of the findings. Additionally, the limitations encountered during the execution of the study will be thoroughly examined in section 5.3, elucidating the constraints and challenges that may have impacted the research outcomes. Finally, the chapter will draw to a close by outlining potential avenues for future research endeavors in section 5.4, offering a succinct overview of areas warranting further exploration and development based on the insights gleaned from the current study.

5.2 Conclusion

This dissertation presents promising outcomes for osteoarthritis (OA) rehabilitation through the utilization of machine learning and computer vision techniques, particularly by leveraging OpenCV to convert webcam images of users and utilizing the Landmark model of MediaPipe Pose for estimating body landmarks. The primary objective of our proposed method is to facilitate precise and efficient identification of exercise posture, thereby minimizing the risk of injury and providing real-time feedback, all of which is cost-effective as this innovation can replace in-person guidance. This program empowers individuals to exercise with correct posture at home, eliminating the need for professional trainer assistance and offering a cost-effective exercise solution. Moreover, the advantage lies in the short time required for posture analysis, allowing users to receive immediate feedback.

While there are numerous technologies available for identifying exercise posture, limitations such as the involvement of domain experts and the high cost of devices hinder widespread adoption by patients. This study proposes a solution utilizing skeleton tracking technology for guidance correction and patient condition tracking, addressing these limitations. As a result, the primary hypothesis—"How effective is the virtual assistant at identifying the position of the body and providing real-time feedback

to correct it during rehabilitation exercises for knee osteoarthritis?"—is affirmed as highly effective.

5.3 Limitations and Constraints

- **Limited Participant Demographics:** The study's participant pool may not accurately represent the broader population, as it primarily included individuals capable of efficiently using laptops, potentially excluding older or less technologically proficient individuals.
- **Lack of Participant Blinding:** Due to the nature of the study design, participants could not be blinded to their intervention, potentially introducing bias in self-reported outcomes.
- **Lack of Reference Values:** The study lacked reference values to differentiate significant improvements in patients, impacting the ability to assess the extent of improvement accurately.
- **Short-Term Evaluation:** The study's evaluation period was relatively short-term, necessitating a long-term follow-up study to ascertain the sustained positive effects of the intervention on exercise adherence and patient health status.
- **Technology Accessibility:** While the virtual physiotherapy guider offers numerous advantages, its effectiveness may be limited by the accessibility and affordability of technology, particularly in regions with limited resources.
- **Need for Further Validation:** The effectiveness and generalizability of the virtual physiotherapy guider need further validation through larger-scale studies involving diverse patient populations and settings.
- **Scope of Outcome Measures:** The study primarily focused on assessing clinical outcomes related to knee osteoarthritis rehabilitation, potentially overlooking other important aspects of patient well-being and treatment efficacy.
- **Used Technology limitations:** One significant limitation of the conducted work is the implementation of the models in Python. While Python facilitated

the rapid development of the test suite, its interpreted nature resulted in slower execution compared to low-level languages like C or C++. Consequently, the testing procedure required a substantial amount of time to run the two models, potentially affecting the overall efficiency and scalability of the study.

5.4 Future enhancements for further research

Future enhancements for further research on this study could encompass several avenues of exploration to address current limitations and expand the scope of inquiry. Some potential areas for improvement and extension include:

- **Performance Optimization:** Investigate strategies to optimize the performance of the implemented models, such as leveraging advanced programming techniques, parallel computing, or utilizing more efficient algorithms to enhance speed and scalability.
- **Integration of Additional Technologies:** Explore the integration of complementary technologies or frameworks that could enhance the functionality and capabilities of the virtual physiotherapy guider, such as collecting and capturing the knee angle measurements by using two or more camera sensors with different angles to accurately measure the knee angle, real-time feedback system and online KOOS questionnaire survey, and incorporating Augmented Reality (AR) for immersive exercise experiences.
- **Personalization and Adaptation:** Develop methods to personalize the exercise guidance and feedback based on individual patient characteristics, such as biomechanical factors, progress over time, pain levels, or adherence patterns, to tailor the rehabilitation program to each patient's specific needs and goals.
- **Long-Term Monitoring and Tracking:** Implement mechanisms for long-term monitoring and tracking of patient progress and outcomes beyond the duration of the study, enabling continuous assessment of the effectiveness of the virtual physiotherapy guider and facilitating data-driven adjustments to the rehabilitation protocol.
- **Integration with Healthcare Systems:** Investigate the integration of the virtual physiotherapy guider into existing healthcare systems or electronic health

records (EHR) platforms to streamline communication and data sharing between patients, clinicians, and other healthcare providers, facilitating seamless coordination of care and comprehensive patient management.

- **User Experience and Engagement:** Conduct user experience research to evaluate the usability, acceptability, and satisfaction of patients and clinicians with the virtual physiotherapy guider, identifying areas for improvement in interface design, interaction flow, and feedback mechanisms to enhance engagement and adherence to the rehabilitation program.
- **Clinical Validation and Comparative Studies:** Conduct larger-scale clinical validation studies to assess the efficacy and comparative effectiveness of the virtual physiotherapy guider against standard rehabilitation protocols or other digital health interventions, incorporating diverse patient populations and clinical settings to evaluate generalizability and real-world impact.
- **Accessibility and Equity:** Address issues of accessibility and equity by ensuring that the virtual physiotherapy guider is designed to be inclusive and accessible to diverse patient populations, including those with limited digital literacy, physical disabilities, or socioeconomic constraints, through features such as multi-lingual support, accessibility features, and affordability considerations.
- **Integration of Behavioral Interventions:** Explore the integration of behavioral interventions or motivational strategies within the virtual physiotherapy guider to enhance patient motivation, adherence, and long-term behavior change, leveraging principles from behavioral science and gamification to promote sustained engagement with the rehabilitation program.
- **Ethical and Privacy Considerations:** Prioritize ethical considerations and patient privacy protections in the design and deployment of the virtual physiotherapy guider, ensuring compliance with relevant regulations and guidelines governing data privacy, informed consent, and ethical conduct in research involving human subjects.

By addressing these future enhancements, researchers can further advance the development and deployment of the virtual physiotherapy guider, contributing to improved outcomes and quality of care for patients undergoing rehabilitation for knee osteoarthritis and other musculoskeletal conditions.

REFERENCES

- Agnes, HajYasien, A., & Kulic, D. (2014). Improving rehabilitation exercise performance through visual guidance. *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*. <https://doi.org/10.1109/embc.2014.6943943>
- American. (2021). *AACN fact sheet - nursing shortage*. [Aacnnursing.org. https://www.aacnnursing.org/news-information/fact-sheets/nursing-shortage](https://www.aacnnursing.org/news-information/fact-sheets/nursing-shortage)
- Arthritis Foundation. (n.d.). *Get in the habit of stretching*. [Www.arthritis.org. https://www.arthritis.org/health-wellness/healthy-living/physical-activity/getting-started/get-in-the-habit-of-stretching](https://www.arthritis.org/health-wellness/healthy-living/physical-activity/getting-started/get-in-the-habit-of-stretching)
- Bazarevsky, V., Grishchenko, I., Raveendran, K., Zhu, T., Zhang, F., & Grundmann, M. (2020). *BlazePose: On-device Real-time Body Pose tracking*. [ArXiv.org. https://arxiv.org/abs/2006.10204](https://arxiv.org/abs/2006.10204)
- Bennell, K. L., & Hinman, R. S. (2011). A review of the clinical evidence for exercise in osteoarthritis of the hip and knee. *Journal of Science and Medicine in Sport*, 14, 4–9. <https://doi.org/10.1016/j.jsams.2010.08.002>
- Bian, S., Rey, V. F., Hevesi, P., & Lukowicz, P. (2019). Passive capacitive based approach for full body gym workout recognition and counting. *2019 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. <https://doi.org/10.1109/percom.2019.8767393>
- Boesch, G. (2022a, June). *A Guide to OpenPose in 2022* - [viso.ai](https://viso.ai/deep-learning/openpose/). <https://viso.ai/deep-learning/openpose/>
- Bora, J., Dehingia, S., Boruah, A., Chetia, Anuraag Anuj, & Gogoi, D. (2023). Realtime Assamese Sign Language Recognition using MediaPipe and Deep Learning. *International Conference on Machine Learning and Data Engineering*, 218, 1384–1393. <https://doi.org/10.1016/j.procs.2023.01.117>
- Boesch, G. (2022b, June). *MediaPipe: Google's open source framework for ML solutions (2022 guide)* - [viso.ai](https://viso.ai/computer-). <https://viso.ai/computer->

vision/mediapipe/

- Buerhaus, P. I. (2008). Current and future state of the US nursing workforce. *JAMA*, 300, 2422. <https://doi.org/10.1001/jama.2008.729>
- Cao, Z., Hidalgo, G., Simon, T., Wei, S.-E., & Sheikh, Y. (2018). *OpenPose: Realtime Multi-Person 2D Pose Estimation using Part Affinity Fields*. ArXiv.org. <https://arxiv.org/abs/1812.08008>
- Carey, K., Abruzzo, B., Lowrance, C., Sturzinger, E., Arnold, R., & Korpela, C. (2020). Comparison of skeleton models and classification accuracy for posture-based threat assessment using deep-learning. *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications II*, 84. <https://doi.org/10.1117/12.2556422>
- Center, N. (2022). Recommendations for the medical management of osteoarthritis of the hip and knee: 2000 update. American College of Rheumatology Subcommittee on Osteoarthritis Guidelines. *Arthritis and Rheumatism*, 43. [https://doi.org/10.1002/1529-0131\(200009\)43:9<1905::AID-ANR1>3.0.CO;2-P](https://doi.org/10.1002/1529-0131(200009)43:9<1905::AID-ANR1>3.0.CO;2-P)
- Centers, & Prevention. (2022). *Osteoarthritis (OA)*. Wwww.cdc.gov. [https://www.cdc.gov/arthritis/basics/osteoarthritis.htm#:~:text=Osteoarthritis%20\(OA\)%20is%20the%20most,underlying%20bone%20begins%20to%20change.](https://www.cdc.gov/arthritis/basics/osteoarthritis.htm#:~:text=Osteoarthritis%20(OA)%20is%20the%20most,underlying%20bone%20begins%20to%20change.)
- Centers, & Prevention. (2023). *Osteoarthritis (OA)*. Wwww.cdc.gov. [https://www.cdc.gov/arthritis/basics/osteoarthritis.htm#:~:text=Osteoarthritis%20\(OA\)%20is%20the%20most,underlying%20bone%20begins%20to%20change.](https://www.cdc.gov/arthritis/basics/osteoarthritis.htm#:~:text=Osteoarthritis%20(OA)%20is%20the%20most,underlying%20bone%20begins%20to%20change.)
- Chang, K., Chen, M. Y., & Canny, J. (2007). Tracking free-weight exercises. *International Conference on Ubiquitous Computing*, 19–37. https://doi.org/10.1007/978-3-540-74853-3_2
- Chen, H., Zheng, X., Huang, H., Liu, C., Wan, Q., & Shang, S. (2019). The effects of a home-based exercise intervention on elderly patients with knee osteoarthritis:

- a quasi-experimental study. *BMC Musculoskeletal Disorders*, 20. <https://doi.org/10.1186/s12891-019-2521-4>
- Chen, J., Caviedes, J., & Li, B. (2021). Classification of single-axis spinal motion using a wearable system of stretch sensors for at-home physical therapy. *2021 43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. <https://doi.org/10.1109/embc46164.2021.9630663>
- Chen, S., & Yang, R. (2018). *Pose trainer: Correcting exercise posture using pose estimation*.
- Chiang, A.-T., Chen, Q., Wang, Y., & Fu, M. R. (2018). Kinect-based in-home exercise system for lymphatic health and lymphedema intervention. *IEEE Journal of Translational Engineering in Health and Medicine*, 6, 1–13. <https://doi.org/10.1109/jtehm.2018.2859992>
- CreakyStaff. (2019, January). *Range-of-motion exercises for arthritis you should do every day*. CreakyJoints. <https://creakyjoints.org/diet-exercise/range-of-motion-exercises-for-arthritis/>
- Cuesta-Vargas, A. I., Galán-Mercant, A., & Williams, J. M. (2010). The use of inertial sensors system for human motion analysis. *Physical Therapy Reviews*, 15, 462–473. <https://doi.org/10.1179/1743288x11y.0000000006>
- DeHaan. (2022). Knee osteoarthritis clinical practice guidelines – how are we doing? *The Journal of Rheumatology*, 34, 2099–2105. <https://pubmed.ncbi.nlm.nih.gov/17722223/>
- Delia, A., & Coroiu, A. (2018). Interchangeability of kinect and orbbec sensors for gesture recognition. *2018 IEEE 14th International Conference on Intelligent Computer Communication and Processing (ICCP)*. <https://doi.org/10.1109/iccp.2018.8516586>
- Deyle, G. D., Henderson, N. E., Matekel, R. L., Ryder, M. G., Garber, M. B., & Allison, S. C. (2000). Effectiveness of manual physical therapy and exercise in osteoarthritis of the knee. *Annals of Internal Medicine*, 132, 173. <https://doi.org/10.7326/0003-4819-132-3-200002010-00002>

- Ding, H., Han, J., Shangguan, L., Xi, W., Jiang, Z., Yang, Z., Zhou, Z., Yang, P., & Zhao, J. (2017). A platform for free-weight exercise monitoring with passive tags. *IEEE Transactions on Mobile Computing*, 16, 3279–3293. <https://doi.org/10.1109/tmc.2017.2691705>
- Doyle, J., Bailey, C., Dromey, B., & Scanail, Cliodhna Ni. (2010). BASE - An interactive technology solution to deliver balance and strength exercises to older adults. *Proceedings of the 4th International ICST Conference on Pervasive Computing Technologies for Healthcare*. <https://doi.org/10.4108/icst.pervasivehealth2010.8881>
- Fang, H.-S., Li, J., Tang, H., Xu, C., Zhu, H., Xiu, Y., Li, Y.-L., & Lu, C. (2022). *AlphaPose: Whole-Body Regional Multi-Person Pose Estimation and Tracking in Real-Time*. ArXiv.org. <https://arxiv.org/abs/2211.03375>
- Farina, D., Jiang, N., Rehbaum, H., Holobar, A., Graimann, B., Dietl, H., & Aszmann, Oskar C. (2014). The extraction of neural information from the surface EMG for the control of upper-limb prostheses: Emerging avenues and challenges. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 22, 797–809. <https://doi.org/10.1109/tnsre.2014.2305111>
- Farooq, A., & Won, C. S. (2015). A survey of human action recognition approaches that use an RGB-D sensor. *IEIE Transactions on Smart Processing and Computing*, 4, 281–290. <https://doi.org/10.5573/ieiespc.2015.4.4.281>
- Fasola, J., & Mataric, Maja J. (2012). Using socially assistive Human–Robot interaction to motivate physical exercise for older adults. *Proceedings of the IEEE*, 100, 2512–2526. <https://doi.org/10.1109/jproc.2012.2200539>
- Felix, Adochiei, Ioana Raluca, Ciucu, R., Pietroiu-Andruseac, G., Argatu, Florin Ciprian, & Jula, N. (2019). Design and implementation of a body posture detection system. *2019 E-Health and Bioengineering Conference (EHB)*. <https://doi.org/10.1109/ehb47216.2019.8969997>
- Fernandes, L., Hagen, K. B., Johannes, B., Andreassen, O., Christensen, P., Conaghan, P. G., Doherty, M., Geenen, R., Hammond, A., Kjekken, I., Stefan, L. L., Lund,

- H., Mallen, C. D., Nava, T., Oliver, S., Pavelka, K., Pitsillidou, I., José, S., Jenny, & Zanolli, G. (2013). EULAR recommendations for the non-pharmacological core management of hip and knee osteoarthritis. *Annals of the Rheumatic Diseases*, 72, 1125–1135. <https://doi.org/10.1136/annrheumdis-2012-202745>
- Goyal, S., & Jain, A. (2021). Yoga pose perfection using deep learning. *Journal of Student Research*, 10. <https://doi.org/10.47611/jsrhs.v10i3.2140>
- Groot, D., Favejee, M., Reijman, M., Verhaar, J., & Terwee, C. (2008). The dutch version of the knee injury and osteoarthritis outcome score: a validation study. *Health and Quality of Life Outcomes*, 6, 16. <https://doi.org/10.1186/1477-7525-6-16>
- Guo, X., Liu, J., & Chen, Y. (2017). FitCoach: Virtual fitness coach empowered by wearable mobile devices. *IEEE INFOCOM 2017 - IEEE Conference on Computer Communications*. <https://doi.org/10.1109/infocom.2017.8057208>
- Hancock, G. E., Hepworth, T., & Wembridge, K. (2018). Accuracy and Reliability of Knee Goniometry Methods. *Journal of Experimental Orthopaedics*, 5(1). <https://doi.org/10.1186/s40634-018-0161-5>
- Health, & Human Services. (2018). *Physical activity guidelines for americans 2nd edition*. Imprints. https://health.gov/sites/default/files/2019-09/Physical_Activity_Guidelines_2nd_edition.pdf
- Health, & Human Services. (2023). *Physical activity and health: A report of the surgeon general*. Centers for Disease Control and Prevention; U.S. Department of Health and Human Services. <https://www.cdc.gov/nccdphp/sgr/index.htm>
- Hochberg, M. C., Altman, R. D., April, K. T., Benkhalti, M., Guyatt, G., McGowan, J., Towheed, T., Welch, V., Wells, G., & Tugwell, P. (2012). American College of Rheumatology 2012 recommendations for the use of nonpharmacologic and pharmacologic therapies in osteoarthritis of the hand, hip, and knee. *Arthritis Care & Research*, 64, 465–474. <https://doi.org/10.1002/acr.21596>

- Hunter, D. J., & Eckstein, F. (2009). Exercise and osteoarthritis. *Journal of Anatomy*, 214, 197–207. <https://doi.org/10.1111/j.1469-7580.2008.01013.x>
- Ichsan Arsyi Putra, Oky Dwi Nurhayati, & Eridani, D. (2022). Human Action Recognition (HAR) Classification Using MediaPipe and Long Short-Term Memory (LSTM). *Teknik Fakultas Teknik Universitas Diponegoro*, 43(2), 190–201. <https://doi.org/10.14710/teknik.v43i2.46439>
- Ishak, N. A., & Addi, M. M. (2017). Rehabilitation exercise monitoring device for knee osteoarthritis patients. *2017 International Conference on Control, Electronics, Renewable Energy and Communications (ICCREC)*, 103–110. <https://doi.org/10.1109/iccerec.2017.8226715>
- Joshi, & Joshi, D. (2021). MoveNet: A deep neural network for joint profile prediction across variable walking speeds and slopes. *IEEE Transactions on Instrumentation and Measurement*, 70, 1–11. <https://doi.org/10.1109/tim.2021.3073720>
- Kang, S. H., Lee, S. J., Press, J. M., & Zhang, L.-Q. (2019). Real-time three-dimensional knee moment estimation in knee osteoarthritis: Toward biodynamic knee osteoarthritis evaluation and training. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 27, 1263–1272. <https://doi.org/10.1109/tnsre.2019.2915812>
- Kang, S. H., Lee, S. J., Ren, Y., & Zhang, L.-Q. (2014). Real-time knee adduction moment feedback training using an elliptical trainer. *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, 22, 334–343. <https://doi.org/10.1109/tnsre.2013.2291203>
- Kang, S. H., Lee, S. J., & Zhang, L.-Q. (2014). Real-time tracking of knee adduction moment in patients with knee osteoarthritis. *Journal of Neuroscience Methods*, 231, 9–17. <https://doi.org/10.1016/j.jneumeth.2013.12.001>
- Ke, Y., ZENG, C., LU, X., & CUI, Y. (2021). Recognition technology of human body movement behavior in fitness exercise based on transfer learning. *2021 6th International Conference on Intelligent Computing and Signal Processing*

(ICSP). <https://doi.org/10.1109/icsp51882.2021.9409004>

Kim, J., Choi, J., Ha, E., & Choi, J. (2023). Human Pose Estimation Using MediaPipe Pose and Optimization Method Based on a Humanoid Model. *Applied Sciences*, 13(4). <https://doi.org/10.3390/app13042700>

Konstantinidis, E. I., Billis, A. S., Mouzakidis, Christos A, Zilidou, V. I., Antoniou, P. E., & Bamidis, Panagiotis D. (2016). Design, implementation, and wide pilot deployment of FitForAll: An easy to use exergaming platform improving physical fitness and life quality of senior citizens. *IEEE Journal of Biomedical and Health Informatics*, 20, 189–200. <https://doi.org/10.1109/jbhi.2014.2378814>

Kothari, S. (2020). Yoga pose classification using deep learning. *SJSU ScholarWorks*. <https://doi.org/https://doi.org/10.31979/etd.rkgu-pc9k>

Lai, D. T. H., Levinger, P., Begg, R. K., Gilleard, W. L., & Palaniswami, M. (2009). Automatic recognition of gait patterns exhibiting patellofemoral pain syndrome using a support vector machine approach. *IEEE Transactions on Information Technology in Biomedicine*, 13, 810–817. <https://doi.org/10.1109/titb.2009.2022927>

Landry, M. (2014). Brukner & Khan's clinical sports medicine. *Physiotherapy Canada*, 66, 109–110. <https://doi.org/10.3138/ptc.66.1.rev2>

Lawrence, R. C., Felson, D. T., Helmick, C. G., Arnold, L. M., Choi, H., Deyo, R. A., Gabriel, S., Hirsch, R., Hochberg, M. C., Hunder, G. G., Jordan, J. M., Katz, J. N., Kremers, H. M., & Wolfe, F. (2007). Estimates of the prevalence of arthritis and other rheumatic conditions in the United States: Part II. *Arthritis & Rheumatism*, 58, 26–35. <https://doi.org/10.1002/art.23176>

Lazar, A., Koehler, C., Tanenbaum, J., & H, N. D. (2015). *Why we use and abandon smart devices*. Proceedings of the 2015 ACM International Joint Conference on Pervasive and Ubiquitous Computing. <https://dl.acm.org/doi/10.1145/2750858.2804288>

Lee, J., Joo, H., Lee, J., & Chee, Y. (2020). Automatic classification of squat posture

- using inertial sensors: Deep learning approach. *Sensors*, 20, 361. <https://doi.org/10.3390/s20020361>
- Li, X., Zhou, Z., Wu, J., & Xiong, Y. (2021). Human posture detection method based on wearable devices. *Journal of Healthcare Engineering*, 2021, 1–8. <https://doi.org/10.1155/2021/8879061>
- Li, Y., Xie, W., Xiao, W., & Dou, D. (2022). Progress in osteoarthritis research by the national natural science foundation of china. *Bone Research*, 10. <https://doi.org/10.1038/s41413-022-00207-y>
- Long, C., Jo, E., & Nam, Y. (2021). Development of a yoga posture coaching system using an interactive display based on transfer learning. *The Journal of Supercomputing*, 78, 5269–5284. <https://doi.org/10.1007/s11227-021-04076-w>
- Lu, T.-W., & Chang, C.-F. (2012). Biomechanics of human movement and its clinical applications. *The Kaohsiung Journal of Medical Sciences*, 28, S13–S25. <https://doi.org/10.1016/j.kjms.2011.08.004>
- Lu, Y., Wang, H., Qi, Y., & Xi, H. (2021). Evaluation of classification performance in human lower limb jump phases of signal correlation information and LSTM models. *Biomedical Signal Processing and Control*, 64, 102279. <https://doi.org/10.1016/j.bspc.2020.102279>
- Ma, J., Ma, L., Ruan, W., Chen, H., & Feng, J. (2022). A wushu posture recognition system based on MediaPipe. *2022 2nd International Conference on Information Technology and Contemporary Sports (TCS)*. <https://doi.org/10.1109/tcs56119.2022.9918744>
- Makarm, Wafaa K, Sharaf, D. M., & Zaghlol, Rabab S. (2021). Impact of home exercise program on self-efficacy and quality of life among primary knee osteoarthritis patients: a randomized controlled clinical study. *Egyptian Rheumatology and Rehabilitation*, 48. <https://doi.org/10.1186/s43166-021-00073-2>
- McAlindon, T. E., Bannuru, R. R., Sullivan, M. C., Arden, N. K., Berenbaum, F., Bierma-Zeinstra, S. M., Hawker, G. A., Henrotin, Y., Hunter, D. J., Kawaguchi, H., Kwoh, K., Lohmander, S., Rannou, F., Roos, E. M., & Underwood, M.

- (2014). OARSI guidelines for the non-surgical management of knee osteoarthritis. *Osteoarthritis and Cartilage*, 22, 363–388. <https://doi.org/10.1016/j.joca.2014.01.003>
- Minor, M. A. (1999). EXERCISE IN THE TREATMENT OF OSTEOARTHRITIS. *Rheumatic Disease Clinics of North America*, 25, 397–415. [https://doi.org/10.1016/s0889-857x\(05\)70075-2](https://doi.org/10.1016/s0889-857x(05)70075-2)
- Nakano, N., Sakura, T., Ueda, K., Omura, L., Kimura, A., Iino, Y., Fukashiro, S., & Yoshioka, S. (2020). Evaluation of 3D markerless motion capture accuracy using OpenPose with multiple video cameras. *Frontiers in Sports and Active Living*, 2. <https://doi.org/10.3389/fspor.2020.00050>
- Nandi, S. (2022). *What is microsoft XNA framework?* C-Sharpcorner.com. <https://www.c-sharpcorner.com/blogs/what-is-microsoft-xna-framework1>
- Nejati, P., Farzinmehr, A., & Moradi-Lakeh, M. (2015). The effect of exercise therapy on knee osteoarthritis: a randomized clinical trial. *Medical Journal of the Islamic Republic of Iran*, 29, 186. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4431424/>
- Nguyen, C., Lefèvre-Colau, M.-M., Poiraudau, S., & Rannou, F. (2016). Rehabilitation (exercise and strength training) and osteoarthritis: A critical narrative review. *Annals of Physical and Rehabilitation Medicine*, 59, 190–195. <https://doi.org/10.1016/j.rehab.2016.02.010>
- NHS Choices. (2019). *Treatment - osteoarthritis*. NHS. <https://www.nhs.uk/conditions/osteoarthritis/treatment/>
- nihr_wp. (2018, August). *People with osteoarthritis can benefit from exercise but may harbour myths about safety*. NIHR Evidence. <https://evidence.nihr.ac.uk/alert/people-with-osteoarthritis-can-benefit-from-exercise-but-may-harbour-myths-about-safety/>
- Norkin, C. C., & Joyce, W. D. (2016). In *Measurement of Joint Motion: A Guide to Goniometry*, 5e (Book, Section). McGrawHill Education. fadvistpt.mhmedical.com/content.aspx?aid=1142184567

- Ofli, F., Kurillo, G., Obdrzalek, S., Bajcsy, R., Jimison, H. B., & Pavel, M. (2016). Design and evaluation of an interactive exercise coaching system for older adults: Lessons learned. *IEEE Journal of Biomedical and Health Informatics*, 20, 201–212. <https://doi.org/10.1109/jbhi.2015.2391671>
- Ono, S., Kusaka, J., Obo, T., & Kubota, N. (2014). Exercise support system with robot partner based on feeling of self-efficacy. *2014 International Symposium on Micro-NanoMechatronics and Human Science (MHS)*. <https://doi.org/10.1109/mhs.2014.7006120>
- Patil, A., Rao, D., Utturwar, K., Shelke, T., & Sarda, E. (2022). Body posture detection and motion tracking using AI for medical exercises and recommendation system. *ITM Web of Conferences*, 44, 03043. <https://doi.org/10.1051/itmconf/20224403043>
- Pattamaset, S., Charoenpong, T., & Charoensiriwath, S. (2020). Evaluation of physical exercise for osteoarthritis of the knee through image processing technique. *2020 12th International Conference on Knowledge and Smart Technology (KST)*. <https://doi.org/10.1109/kst48564.2020.9059534>
- Peilei, Y. (2021). *Effect of systematic exercise rehabilitation on patients with knee osteoarthritis: A randomized controlled trial - jing chao, zhang jing, bai xuehua, yang peilei, gong qi, 2021. CARTILAGE.* <https://journals.sagepub.com/doi/full/10.1177/1947603520903443>
- Pendleton, A. (2000). EULAR recommendations for the management of knee osteoarthritis: report of a task force of the Standing Committee for International Clinical Studies Including Therapeutic Trials (ESCISIT). *Annals of the Rheumatic Diseases*, 59, 936–944. <https://doi.org/10.1136/ard.59.12.936>
- Pereira, L. C., Rwkabayiza, S., Lécureux, E., & Jolles, B. M. (2015). The knee smartphone-application goniometer is a more reliable tool than the standard goniometer in acute orthopaedic settings. *Physiotherapy*, 101, e1192–e1193. <https://doi.org/10.1016/j.physio.2015.03.2117>
- Pueo, B. (2016). High speed cameras for motion analysis in sports science. *Journal of*

Human Sport and Exercise, 11, 53–73.
<https://doi.org/10.14198/jhse.2016.111.05>

Rahman, A., Qamar, A. M., Ahmed, M. A., Rahman, A., & Basalamah, S. (2013). Multimedia interactive therapy environment for children having physical disabilities. *Proceedings of the 3rd ACM Conference on International Conference on Multimedia Retrieval*, 313–314.
<https://doi.org/10.1145/2461466.2461522>

Ruane, J. J. (2011, September). *Exercise to manage knee osteoarthritis*. Practicalpainmanagement.com; PPM.
<https://patient.practicalpainmanagement.com/conditions/osteoarthritis/knee/exercise-manage-knee-osteoarthritis>

Sandmeier, R. H. (2000). Osteoarthritis and exercise: Does increased activity wear out joints? *The Permanente Journal*, 4, 26–28.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6220608/>

Shotton, J., Fitzgibbon, A., Cook, M., Sharp, T., Finocchio, M., Moore, R., Kipman, A., & Blake, A. (2011). Real-time human pose recognition in parts from single depth images. *CVPR 2011*. <https://doi.org/10.1109/cvpr.2011.5995316>

Susan Bartlett. (2018, January). *Role of exercise in arthritis management*. Johns Hopkins Arthritis Center. <https://www.hopkinsarthritis.org/patient-corner/disease-management/role-of-exercise-in-arthritis-management/>

Teng, G., Hong, F., Xu, Y., Qi, J., Jiang, R., Liu, C., & Guo, Z. (2020). MobiFit: Contactless fitness assistant for freehand exercises using just one cellular signal receiver. *2020 16th International Conference on Mobility, Sensing and Networking (MSN)*. <https://doi.org/10.1109/msn50589.2020.00057>

TensorFlow.js. (2021). *Next-generation pose detection with MoveNet and TensorFlow.js*. Tensorflow.org. <https://blog.tensorflow.org/2021/05/next-generation-pose-detection-with-movenet-and-tensorflowjs.html>

Thiengwittayaporn, S., Wattanapreechanon, P., Sakon, P., Peethong, A., Ratisoontorn, N., Charoenphandhu, N., & Charoensiriwath, S. (2021). Development of a

mobile application to improve exercise accuracy and quality of life in knee osteoarthritis patients: a randomized controlled trial. *Archives of Orthopaedic and Trauma Surgery*. <https://doi.org/10.1007/s00402-021-04149-8>

Tolentino, R. E., Mae, & Ysabelle, D. (2019). Recognition of different emergency situation through body gesture using microsoft kinect sensor. *2019 IEEE 11th International Conference on Humanoid, Nanotechnology, Information Technology, Communication and Control, Environment, and Management (HNICEM)*. <https://doi.org/10.1109/hnicem48295.2019.9073365>

US Census Bureau. (2022, January). *International database*. Census.gov. <https://www.census.gov/programs-surveys/international-programs/about/idb.html>

Wang, C.-Y., Bochkovski, A., & Liao, H.-Y. M. (2022). *YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors*. ArXiv.org. <https://arxiv.org/abs/2207.02696>

Webborn, A D J. (1997). ACSM's exercise management for persons with chronic diseases and disabilities. *British Journal of Sports Medicine*, 31, 354. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1332589/>

Wiedemann, L. G., Kampel, M., Planinc, R., & Nemec, I. (2015). Performance evaluation of joint angles obtained by the Kinect v2. *IET International Conference on Technologies for Active and Assisted Living (TechAAL)*. <https://doi.org/10.1049/ic.2015.0142>

Wrigley, T. V., Simic, M., Hunt, M. A., Hinman, R. S., & Bennell, K. L. (2009). Real-time movement biofeedback for walking gait modification in knee osteoarthritis. *2009 Virtual Rehabilitation International Conference*, 132–135. <https://doi.org/10.1109/icvr.2009.5174219>

Zeng, C.-Y., Zhang, Z.-R., Tang, Z.-M., & Hua, F.-Z. (2021). Benefits and mechanisms of exercise training for knee osteoarthritis. *Frontiers in Physiology*, 12. <https://doi.org/10.3389/fphys.2021.794062>

Zhang, W., Moskowitz, R. W., Nuki, G., Abramson, S., Altman, R. D., Arden, N.,

Bierma-Zeinstra, S., Brandt, K. D., Croft, P., Doherty, M., Dougados, M., Hochberg, M., Hunter, D. J., Kwoh, K., Lohmander, L. S., & Tugwell, P. (2008). OARSI recommendations for the management of hip and knee osteoarthritis, Part II: OARSI evidence-based, expert consensus guidelines. *Osteoarthritis and Cartilage*, 16, 137–162. <https://doi.org/10.1016/j.joca.2007.12.013>

APPENDICES

